RANAYAMA



BREATHING FOR BETTER HEALTH

IN THE ACT OF BREATHING, THERE SHOULD BE A RHYTHM OF REST AND WORK. NOT THE LUNGS ALONE, BUT NEARLY THE WHOLE BODY, IS MADE ACTIVE BY RIGHT BREATHING. THIS ACTIVITY IS NECESSARY FOR HEALTH.

K. LAKSHMANA SARMA

"Health is the basis of Yoga and perfection. This Pranayama dealt with in this book, though not the traditional Pranayama of yogis, nevertheless helps to lay the basis of sound good-health which is the gateway to the higher attainments of yoga. Many secrets of correct breathing and its connected requirements are revealed here....."

Swami
Chidananda Sarasvati
President
Divine Life Society
Rishikesh

With compliments
from
Dr. H. Sreenivasiah
L Tyagaraja Sharma
27.3.74

ಜಯಚ್ಚೀ ಟ್ರಸ್ಟ್
ಪರಿಷತ್ ನಿಲಯ
ಕ್ರಮಸಂಖ್ಯೆ 1885
ವಿಭಾಗ ..... ಸಂಖ್ಯೆ 0.2

ACC. No. CL 4609
No. 0019 - 4609
DATE

Printed at
SARMA'S SANATORIUM PRESS
Ganesanagar
PUDUKKOTTAI—1. Tamil Nadu

PRANAYAMA
OR
BREATHING FOR BETTER HEALTH

K. LAKSHMANA SARMA
Founder-Director, The Indian Institute Of Natural Therapeutics

Revised by
L. GANESA SARMA
Secretary, The Indian Institute Of Natural Therapeutics

Sixth Edition

February, 1974

Published by
THE INDIAN INSTITUTE OF NATURAL THERAPEUTICS
1728 (G–168), Nanakpur, New Delhi–110021

SOLE DISTRIBUTORS
THE NATURE-CURE PUBLISHING HOUSE
Ganesanagar, Pudukkottai–622001

Rs. 6 60 pence 2 dollars

CONTENTS

		Page
	Foreword by Swami Chidananda Sarasvati	5
	Preface to the Fifth Edition	7
	Preface to the Sixth Edition	8
	Introduction	9
I.	The Bases Of Life	13
II.	Dress and Health	18
III.	The Air	29
IV.	Physiology Of Respiration	35
V.	Practice Of Breathing	70
VI.	Notes on Sleep	96
	Appendices	
I.	Cure Of Asthma through the Life Natural	103
II.	From The Vedic Lore	111
	Important Note	121
III.	From Chapter XIV, Prakeernakam, of the Samskrit book, Svaadheena - Svaasthya - Mahaa-vidya, by the Author	
	1, Vastradhaarana (Dress)	127
	2. Praanaayaama (Breathing For Better Health)	130
V.	On Applying Oil Over the Body	137
	Index	142

ILLUSTRATIONS

		Page
1.	Trousers suspended by shoulder braces keep it in position without pressure on the abdomen	25
2.	Rolling the dhoti down the chest	27
3.	Rolled down dhoti -- loose, but not slipping down	
4.	The position of the lungs in the structure of the human body	35
5.	The thorax	36
6.	The Ciliated cells	38
7.	The nasal passage	39
8.	The larynx	42
9.	The vocal cords -- enlarged	
10.	The trachea (front view)	44
11.	The bronchi	45
12.	The lungs	47
13.	The lungs showing the alveoli	48
14.	The lobes of the lungs and the pleura	51
15.	The organs of the chest -- lungs, heart, etc.	54
16.	The chest bones	55
17.	The heart -- arteries, veins	68
18.	Exhaling residual air through the mouth	81
19.	The wrong way -- the abdomen accommodating for inspiration and expiration, the chest remaining rigid	83
20,	Effect of inspiration and expiration on the chest and the diaphragm -- lateral view	84
21.	Effect of expiration on the lungs, chest and diaphragm -- front view	85
22.	Effect of inspiration on the lungs, chest and diaphragm -- front view	
23.	The chest in normal expiration	85
24.	The chest in normal inspiration	
25.	Proper sitting posture for the practice of Pranayama	90
26.	Pooraka through the left nostril	91
27.	Baahya-Kumbhaka	
28.	Pooraka through the right nostril	92
29.	Aantarik-Kumbhaka	
30.	The mudra -- the mode of holding the right hand for the practice of Pranayama	93

FOREWORD

Homage unto the Divine !

Salutations to the worshipful Guru! May the Grace of Sri Anjaneya, the celestial offspring of Sri Vayu Deva confer upon you long life, sound good health, strength and happiness. Success and Victory attend upon you!

It gives me great joy to write these few lines to invoke the Grace of the Lord and the choicest blessings of Sri Sadguru Swami Sivananda upon this excellent and invaluable publication, "PRANAYAMA" or Breathing for Better Health. This is its Sixth Edition and is a distinct improvement upon previous editions in that it is enriched by nearly thirty valuable illustrations, extremely instructive and educational. These improvements are the results of the earnest and dedicated labours of the very devoted son of the late revered author of this book.

This is a unique book which has been especially written for the benefit of people in ill-health who are striving to regain normal health. While Water, Sun-light and Earth are employed successfully in the process of Naturo-therapeutics, the use of two subtler elements, Air and Ether (Akasha) is not so knowledgeably utilised. The wise utilisation of the ebb and flow of Vayu and Akasha within the respiratory

area of the human organism is the vital art scientifically explained and expounded in this little book. Health is the basis of yoga and perfection. This pranayama dealt with within this book, though not the traditional pranayama of the yogis, nevertheless helps to lay the basis of sound good health which is the gateway to the higher attainments of yoga. Many secrets of correct breathing and its connected requirements are revealed here from out of the personal experience of the worshipful author and his profound inner reflections upon this subject of vital importance to human welfare and happiness.

The publishers have rendered a signal service to the peoples of Bharata Varsha in preparing and bringing out this present edition in the cause of National Health. I wish this book the widest possible circulation and reading by the public. Being the work of a sage and spiritual personality its pages carry with them his benedictions.

I wish the Divine Grace of the Almighty and the blessings of Gurudev upon all the readers. May you attain Radiant Health and Eternal joy! Om Shanti!

Sivananda Ashram,
Rishikesh
July 31, 1973.

Swami Chidananda.

PREFACE

To The Fifth Edition

Pranayama, as ordinarily understood, is a strenuous process needing to be practised under the personal supervision of an expert by one who is already free from disease. But intending followers of Nature-Cure are far from being healthy enough for this practice. For them a perfectly non-violent method is necessary. Such a method is the one that is set forth in this booklet. Being non-violent, it is safe for even patients to practise at home without supervision by an expert.

The mode of practice recommended here is intended as an aid to health and nothing more. It is therefore different from Yogic Pranayama. It is only barely indicated in the following pages. But even the modified Pranayama described and recommended in these pages is quite sufficient for followers of Nature-Cure for progressive improvement of both bodily and mental health.

Camp
H.J. Nature-Cure Hospital
Malad (Bombay)

K. Lakshmana Sarma
1-2-1949

PREFACE

To The Sixth Edition

At the time of his stay with me in New Delhi in the year 1963, which, Father said at the time of his leaving for Pudukkottai, will be the last visit, I had the greatest privilege of discussing, among many other matters, the need to revise some of his books that have gone out of print, which included this one, and also of the topics to be included, modified or improved.

After a lengthy exchange of thoughts, he was all grace to say that, because he found that my ideas on the various aspects of the subject were firmly established on the basic principles expounded by him in his "Practical Nature-Cure" and "Svaadheena-Svaasthya-Mahaavidya" (Samskrit), I should undertake to revise his works.

In revising this one, I have just commenced to implement his loving commands. Though it is almost impossible for me ever to swerve from the firm track laid down by him for us, I have taken every care to check up every aspect of the subject, before it went to print.

A proof of this fact was had when I wrote my first book, "Spinal Bath For Nervous Tone". I had the Divine blessings of having it scrutinized by Father and obtaining his approval in that it needed little modification and also his firm conviction that my opinions were just a reflection of his own.

I lay at His Holy Feet this Flower of my first attempt of revising Father's work. May he bless all his followers —old and new—with all that is good.

L. Ganesa Sarma

INTRODUCTION

By L. Ganesa Sarma

The word Pranayama is a compound one which can be split as Praana—aayaama: (प्राण—आयाम:) which means stopping of the breath in between respiration while reciting the sacred mantras. In this revised edition of this book the last aspect is also dealt with, yet, we believe, instructions in mantras and the proper manner of their use are only to be had directly from a chosen master (गुरु).

Most masters of Pranayama do only stress the stoppage of breath after inhaling. Father has been the discoverer of quite many a therapeutic measures, not so much after a scientific exploration, but as guided by his own intuition. And his intuition did work wonders when necessity presented problems before him. I could never say that he was confronted by any problem, for quick was his brain wave and he always had the solution almost ready.

His intuition had never run amock. It could not, for he was firm about the basic principles that he had expounded and they are unshakable, because they were based on axiomatic philosophic Truths.

Pranayama is normally considered to be an exercise. Ease loving people do not naturally take to it. They do not like to strain themselves. People, who are for no work, are those who lack something vital—Health. They therefore need what is commonly called rejuvenation. The need has arisen because most of the vital organs have been overworked and the lungs are among them.

It is a mistake that they have been overworked and the remedial measure should naturally be one that should make amends for the mistake committed earlier. That means that the lungs need *resting*.

But Pranayama is an exercise, which will mean adding strain to the strained organs; and that will not therefore improve the situation.

The lungs have the vital duty of supplying Praana and eliminating Apaana. This has not been so far done properly, and as a result the whole blood stream has been fouled, causing deposits of morbid matter in various parts of the body. The need therefore is also to improve the manner of breathing. That requires a change from wrong to the right. Such a change has to be achieved by a course of training on proper lines. Therefore, that will need Pranayama.

On the one side we reject Pranayama as a strenuous task and on the other we need it as a right course of training. Father therefore thought that if a way be found whereby there can be a harmonious combination of the two, namely work and rest, that will be *the real solution* to the problem.

The lungs are active in all the three steps of Pranayama, viz., inhaling, retention and exhaling. These provide exercise. Yogis generally consider the act of retention of the breath as a period of rest. But it did not so much appeal to Father's mind. He was aware that when the air is held inside the lungs there is work of exchange of gases. Even taking it to be partly a period of rest, for the interchange of gases is

over in quite a short time and the period of retention is normally longer, there is no rhythm of work and rest in a continous cycle. In as much as inhaling, retention, and exhaling, and again inhaling, retention and so on, the order becomes a repetition of one step of work, one step of rest and one step of work. The first and the last steps being of the same kind, i. e., work, the continuity of cycles will be thus: Work, rest, work; work, rest, work; work, rest, work and so on. Here, therefore, we do not have a rhythm of work: and rest. Even the rest that is there is only a supposition. It would be really rhythmic if the cycle were thus: one step of work, one step of rest, one step of work and one step of rest; thus forming an even number of steps to complete one cycle, so that the end of one cycle will have no duplication of the same type of step with the beginning of the next cycle, and also keep up the rhythm in harmony.

This therefore drove Father to the conclusion that introducing a pause after exhaling, to break two steps of work going together, viz., exhaling and inhaling. It was not difficult for him to give it a name in the traditional manner—Baahya-Kumbhaka.

He had done this with just an elementary knowledge of the exercise, keeping more in mind the need for utmost safety for any one who practises it, whether a learner or an advanced adept. As far as he was concerned, therefore this was an independent discovery. But I do not claim that he was the first to tell it to the world. Much later than his discovery, a certain gentleman pointed out that this was already a limb of the Pranayama as explained by a certain ancient author, and wonderfully enough, the name for that particular step is exactly the same. But yet the gentleman

did agree that the reason for which Father brought in this one seems to be his original idea.

Father did not stop with that. To deserve rest, one must have done some work. Another thing is to fill in something valuable, a clean place should be found. Therefore he thought it fit to exhale first, before resting and inhaling after that. He also added that the extent of filling depends upon the extent of emptiness created. Therefore, he stressed the greatest importance of emptying the lungs as fully as one can; if concentration be on that aspect, the rest will automatically follow suit. Keeping all these reasons in mind, he has given Rechaka the first place,

We are not to take the literal meaning of the word 'Pranayama', but that which it implies; and that is indicated in the sub-title of the book; "Breathing For Better Health." Though the lungs form the major breathing organ, it cannot be denied that we breathe through our skin also. Therefore maintenanc eof skin health is also a subject of this book.

We breathe while we are awake and while asleep alike. It is therefore essential to study how best to sleep to the best advantage of the health-seeker. Naturally, that has also a place in this book.

Every attempt has been made to provide e nough technical knowledge of the physiology of the lungs, so that our students are in a position to understand the why and wherefore of every lesson that is taught in this book.

———

1. THE BASES OF LIFE

The central lesson of Life is this. Our well-being, both here and hereafter, depends upon the abundance of our vitality. As Milton said, "To be weak is miserable, doing or suffering". It is the vital power that carries out all the functions of the body, which includes the involuntary ones, whether we are aware of them or not. The involuntary functions can be named as metabolism with its two branches, viz., anabolism and catabolism (in simple language, assimilation and elimination). What is left over after all these are served should alone be utilised for the conscious activities of the worldly life. A reckless drain of the vital power has caused a rapid depletion of the Vital Reserve, and a judicious and economical utilisation helps to maintain a good reserve of Vital Energy[1]. This alone is the source and support of health and a happy frame of mind.

---

1 Priority should be given to the vital labours stated above and not the conscious activities, and it is the reversal of this natural order that results in disease through failure of Vital Economy. The practical implications of this principle are fully dealt with in the author's book, "Fasting Cure & Vital Economy".

Vital power comes to us from cosmic store, from Nature. What we need to do, to be getting ample supplies of it, is to keep up our communion with Her in all Her forms, of which five are mentioned, namely Ether (आकाश), Air (वायु), Light (तेजस्), or Fire, Water, (अप्), and Earth (पृथ्वी). The last named, i. e., Earth, should not be taken to mean mud, clay or dust, but food from plants that grow on earth. The details of this lesson, drawn from the Taittiriyopanishad is given in the "Practical Nature-Cure", by the author.

The above is the natural order of these forces, according to their values for life, health and happiness. To use them in accordance with this order is the only way to maintain the Rhythm of Life. The reason is that every one of these five elements can bestow the full benefit of its utilisation only if the foregoing, elements are given their due priority. To make this point intelligible, we could say that fire can exist only if the foregoing, viz., Ether and Air are also there; similarly earthy mattter, i. e., food, can exist only if all the four elements elder to it are present. Man's communion with Ether must be constant--unbroken. His communion with Air ought to be equally so. He needs less of light and still less of water. The

least of all our vital needs is food from Earth. Our present ways are quite contrary to this sacred law of Nature. And this is the reason for all our ill-health and premature old age and death.

Among these five sources, the three higher ones may be regarded as food for life, and the latter ones as food for the body. Life is an independent entity, separate from food and superior to the body.

Of the higher three, Ether is the first, being the finest and the eldest in creation. The Vedanta says: "Who can live or breathe if this blessed Ether were not?"

Ether communion is kept up by the rhythm of work and rest, digestion and other labours. Its practical expression is the rule, "Be hungry", which means that we should never eat all we can manage to fill in, let alone all we can digest, but much less, so as to leave enough power for other uses, and for economical utilisation of the Vital Reserve. The same thing has been said in other words by Edward Earl Purinton: "The secret of being Healthy, Holy and Happy is to be a little hungry most of the time". Unceasing natural joy—joy uncaused by

outside objects–is the reward of Ether–communion, as observed by the famed Tamil Saint-Poet, Tiruvalluvar, the author of the "Kural":

பொறிவாயில் ஐந்தவித்தான் பொய்தீர் ஒழுக்க
நெறிநின்றார் நீடுவாழ் வார்."

2 "पंचेन्द्रिय-निग्रकीये, प्रभु का किया विधान ।
धर्म पंथ के जो पथिक हों चिर आयुष्मान् ॥

"He that has controlled the five senses, and follows the Path of Righteousness, will live in peace for long."

Food contains medicinal elements, which require knowledge and care so that they may be conserved. When these elements are lost through cooking or processing, then food does not really nourish even if heavy meals are taken, and disease is inevitable. This dietetic science is practically a sealed book to the rank and file of the medical profession. Intelligent laymen should therefore study the subject for themselves.

Water-Cure, one of the branches of Nature-Cure, has obtained greater vogue than it deserves. Where there is a poverty of the

2 From a Hindi translation of the original Tirukkural, by Sri Venkatakrishnan and published by Sri Valmikanatha Pillai for the Trukkural Prachara Sangh, 3, Annamalaipuram, Tiruchirappalli (Tamil Nadu)

higher elements in the body, there Water-Cure is bound to fail, if it is alone relied upon. Air–Cure and Sun–Cure are infinitely far more valuable.

Thus it can be said that there are five indispensable elements that form the constituents of a rational Natural Food-Medicine, namely Ether, Air, Light, Water and Earth in the form of gross food from plants that grow on Earth. In serious cases the first four of these five are necessary and neglect of one or more of these may lead to failure.

2. DRESS & HEALTH

Our supply of air, as well as of light, is the less, the more unnaturally we dress ourselves.

The cult of nudity, now being practised by a few in Germany and elsewhere, ought to give us furiously to think. In our eagerness to be thought decent folk by foreigners, we have gone far away from our ancient customs in dress. We carry too much weight in clothing and smother our skins to death.

For the sake of our skin-health, we ought to practise nudity, or an approach to it whenever it is possible, in private, exposing the whole skin surface to God's light and air.

Very little interchange of gases occurs by the skin in human beings. If we lose daily about 900 grams of carbon-di-oxide by means of the lungs, we excrete only about 9 grams through the skin. Naturally, the oxygen absorbed through the skin is also less. However the skin is an important means of getting rid of water, inorganic salts (including common salt consumed in diet), and traces of urea. It is a matter never to be forgotten that the skin

compensates for the lack of elimination of urea etc., by the kidneys, whenever it is weak or affected. Such elimination is of 'considerable importance because such elimination of inorganic matter also includes drugs put into the system through any means. The skin also absorbs matter applied over it. This property is utilised in skin treatment. Therefore application of ointments and other poisonous preparations by a follower of the Life Natural is improper.

Calling themselves naturopaths, some people use skin applications and when questioned, explain away their mistake by saying that the skin cannot absorb directly, since it is not a digestiva organ. If they believe in this, why should they waste money and time on such applications? More information on this subject can be seen in the Appendix at the end of the book.

The amount of water lost by the skin daily varies between 500 and 200 cubic centimetres. The factors that govern the amount of excretion of sweat, visible or invisible or insensible are (1) the level of external temperature, (2) the amount of muscular activity and (3) the content of inorganic wastes in the system.

With a high temperature the skin capillaries undergo dilation and with a lower temperature they undergo constriction and obviously there is absence of visible sweat, But a severe psychic shock is also followed by a profuse pouring out of sweat. This is known as nervous sweat, because the sweat glands are stimulated by nerves derived from the sympathetic system, which is quickly thrown into action by fear. Sweating because of a high temperature and sweating because of psychic reasons is different in that in the latter the skin becomes pale. A similar condition of nervous sweating can also occur in a state of nausea.

In addition to forming of sweat, the skin also secretes sebum, a natural lubricant prepared by the sebaceous glands. When the fine ducts of these glands are blocked up, with dirt and other material, such as talcum powder, they give rise to blackheads, such as pimples and (summer) boils. Attempts to "cure" them with ointments, etc. may cause the running together of a number of boils, involving a considerable area of the subcutaneous tissue (tissue just below the skin), and form a carbuncle. *Suppression of skin troubles, such as eczema, will run back inward*

and affect the bigger breathing organ, the lungs, causing asthma.

Pigmentation of the skin provides a protective mechanism, being thus able to stand ultra-violet irradiation which will otherwise cause irritation, inflammation and even ulcerations. Therefore a dark skinned person is at an advantage.

Sunlight (its ultra-violet rays) acting upon the skin converts the fat secreted by the sebaceous glands into vitamin D. This vitamin, just like vitamin A, is also fat soluble. Therefore, if that fatty substance is washed off by the use of soap, the body cannot get vitamin D. Natural washing substances such as certain leaf powders prepared after drying the leaves in the shade (உசிலந்தழை), Bengal or green gram flour or shikaikai beans boiled and ground into a paste, preferably with fenugreek (methi) seeds or shikaikkai powder (शिकैकाय) can be used for the maintenance of cleanliness of the skin.

Vitamin D is quite essential for the metabolism of both calcium and phosphorus and therefore calcium phosphate which is the main mineral salt in the bones and teeth cannot be formed, resulting in softening of

bones. Lack of vitamin D in children causes rickets. Pregnant and suckling mothers need an abundant supply of this vitamin as it is essential for the growth of both the foetus and the child. Calciferol, a commercial preparation of vitamin D is different in chemical structure and composition, as is opined by authorities and hence cannot be of help, besides the reason that anything inorganic cannot form part of the body.

Through the sweating mechanism, the skin is able to regulate the body temperature, whether the person is in the poles or working as a stoker in a railway engine in the tropics. But that will be possible only if the skin be covered with a porous material. The difference between a porous dress and an absolutely waterproof texine provided in bus and railway seats can be experienced very well on a summer midday. That portion of the body that is covered by the non-porous material is as good as covered with a thick layer of paint. Kenneth Walker, in his book "Human Physiology" cites the example of a certain boy who took part in a papal procession, and whose body was gilded all over in order that he might represent a golden cherub, and who died the same night, because of interference

with the heat regulating mechanism of the skin.

The same effect, in a lesser degree, is being experienced by the skin of those who use close textured and artificial fabrics, such as nylon, terylene, etc. Urticaria and other skin troubles have been noticed to have been caused by wearers of artificial clothing material. Skin specialists have warned many such sufferers against the use of such material.

Wrong dress spoils health in another way also. It shuts *in* the foul gases that are given off through the skin, and they are breathed in again through the same medium.

Those who wear the least dress have the best skin-health. An American Indian was questioned how it was that he enjoyed a splendid skin-health; he replied in his broken English, "My body all face".

The quality as well as the quantity of the cloth used and the mode of wearing are all important. As civilisation advances, ladies progressively reduce their clothings and gents make good for all that and at the same time make them skin tight.

Coarse, porous and absorbant, cotton clothing of the type of khaddar, are alone suitable for use during hot weather and natural woollen fabrics for the winter.

We wear much dress, much of it for the admiration of others than for utility. While at home we do not need to wear anything more than our dhoties or pyjamas. It is inexcusable to wear shoes of European pattern, with socks (even that of nylon and such artificial material) also during the hot weather. In most part of India sandals or chappals which permit passage of air is quite sufficient. It is not a matter for mockery, if one can go barefooted in all weathers. Such people should only take care to keep their feet also clean. That is the main reason for the stipulation of washing the feet before entering any premises. It is doubly beneficial in that it permits of contact with the hot ground or the soothing grassy lawns in between sometimes and cooling by washing to relieve the tiresomeness in the feet.

Then, as to the mode of wearing, we have fallen into habits involving grievous violence to the vital organs inside the belly and the chest. Owing to long and continuous use there may be no feeling of strain or pressure.

But that is because the organs have got displaced and fixed in wrong positions, whose mischievous effects are wholly chronic. Of course the breathing as well as the digestion is affected by these displacements.



Figure 1.
Trousers suspended by shoulder braces keep in position without pressure on the abdomen.

Those who wear trousers crucify their internal organs by tightly worn belts. They should use proper shoulder braces, and discard belts.

Even so, the dhoti wearers tighten the upper edge of the cloth around the waist, at the navel level, digging deep into the flesh and forming an ugly and permanent groove all round the waist. Ladies using silk and other close textured fabrics develop skin eruptions or eczema about that region.

Perhaps it would be necessary for them to attend a class where they can be taught how to wear the dhoti in a non-violent way. I shall however try to convey to the reader as clear an idea as possible of the right method.

In this method the dhoti, when worn will have its upper edge resting quite loosely, not at the navel-level, but two or three inches (five or eight centimetres) lower like a loose girdle. Even in the deepest deep breathing, in which the abdomen is pushed down and bulged outwards, this girdle will not exert any pressure on the abdomen.

How to adjust that dhoti then? Usually a length of a yard or more on the left, and a longer length on the right, are allowed, to be tucked up in folds behind and in front, as every one knows, and the middle portion of the upper edge is drawn tightly around the waist, the cloth on the right side overlapping that on the left is folded and tucked up in front. The cloth hanging down on the left is passed between the legs, taken to the back side and tucked up after suitably folding the length. That on the right side, as said earlier, is tucked in after folding along the lengthside, while that in the rear along the width side. Usually

the width side is also folded along the width side also to show off graceful folds; but some prefer to avoid the second tucking up in front. This way the dhoti has graceful folds and at the same time permits of sufficient ventilation. This is the usual way.



Figure 2. Rolling the dhoti down the chest.
Figure 3. Rolled down dhoti---loose, but not slipping down even though worn daylong.

The reformed way is also just the same with just an initial difference in fixing the top edge of the cloth around the waist. After placing the two sides, right over the left

around the waist at a level with the nipples of the chest, (it is supposed that the chest will be wider than the belly), the two layers of cloth are rolled down together (see picture -Figure 2). It will be a loose girdle, but will never slip off, even if it be worn for the whole day.

There is fear that if one wears his dhoti loose one might develop a pot-belly. The opposite is the case. Those who wear tight clothes, prevent the movement of gases in the belly along the intestines whose course crosses the girdling cloth both up and down and tends to bloat, resulting in a pot belly, not easy to shrink again. Therefore, those who are for a good form should take care to be kind to the belly, not to do self-violence and get an ugly hanging belly.

For the first few days there will be a queer feeling when dress is worn loose, be it the dhoti or the pants with braces; but that will pass of if it be disregarded.

———

3. THE AIR

It is a common saying that we are air-animals. We should rather say "open-air animals". Well or ill, we breathe all the time, day and night, awake and asleep. If we cease to breathe for more than a few minutes, we cannot live; we would pass out of our bodies. Our need of sun-light, water and food is certainly much less. As explained earlier, Air is the second of the three finer Elements. We can go without food for many weeks at a time. If we do not get air enough, the Rhythm of life would be lost.

In the act of breathing, there should be also a rhythm of rest and work. Rhythmical breathing allows of rest to the breathing machinery in every single breath and thus makes long life possible.

By right use of the body in breathing, it is also kept in due exercise. Not the lungs alone, but nearly the whole body, is made active by right breathing. This activity is necessary for health. Inactive organs are forced to receive the seeds of disease, in so far as they are not eliminated. Very poor breathing is one of the causes of tubercular degeneration of the lungs.

Are we getting enough air? How much air do we need? And where can we get the *kind* of air needed?

Among the more enlightened members of the allopathic profession, there is one named Dr. Leonard Williams, whose book, "The Science And Art Of Living", is worthy of study. According to him the doctor world is going astray on questions of hygiene, it is still engaged in finding the **minimum** of our needs –in vitamins, light and air–without which we shall, in a few weeks or months, contract some fatal disease such as beri–beri. This betrays an utterly vicious outlook.

When even the doctorially estimated minimum is denied, disease follows so quickly that the causal relation is clear. But it does not follow that this standard confers a positive health and life-long immunity from disease. The profession is yet to realise the mischief of **slow** starvation (starvation is caused by denying any one or more of the primary foods of a living being, viz, the Five–fold Food Elements, explained earlier) which manifests itself after many decades of erroneous living. Even a very slight deficiency in the course of years must accumulate till it becomes serious. It is the

maximum of these vital needs that we should seek to obtain to be on the safe side. Any minimum is purely fanciful. There can be no over–consumption of these finer elements especially of Air or Ether.

The maximum of air supply we can get by an open air life, and also by better way of breathing, and more correct ways of dress (explained in the earlier chapter), because we breathe even through the skin, though not as much as through the nostrils.

From the air that we breathe in, it is only oxygen that is utilised for the purpose of oxi–dation of food essences to produce heat. This need for oxygen can be greatly reduced by minimising the food intake, bringing it down to the necessary limits than the possible prey[3]. Dietetic righteousness is therefore necessary for efficient breathing also, because if the belly

3 Thg word prey is used to denote ravenous eating. If one preys on food, one also becomes a prey to ill-health. This has also been said by the Sage Tiruvalluvar:

இழிவறிந்து உண்பான்கண் இன்பம்போல் நிற்கும்
கழிபேர் இரையான்கண் நோய்.

भला समझ मित भोज का जीमे तो सुख-वास ।
वैसे टिकता रोग है अति पेटू के पास ॥

Tirukkural (Hindi) 946

is filled to capacity the space available for the lungs to expand will be far limited than when it is reasonably occupied with food.

An open air life may possibly make it unnecessary to practise better breathing, especially if one also sleeps in the open air. This sleeping in the open, or nearly so, has been found possible even in cold seasons, within the limits of non-violence. Blankets or quilts can be used for the sake of warmth of the whole body and even the head, but not the face. If it becomes absolutely necessary to sleep inside a shelter, it must be as much open as an open space except for the top. Otherwise, it must at least be a place where cross ventilation is possible. Breeze does not blow always from the same direction. There must be windows enough openable in all directions so that air can enter from one direction and get out through the opposite one. People sleeping should choose their places in between that course of the breeze.

Even if one practises better breathing, but indoors, where the air is not thus renewed often, he would be doing himself more harm than good: for the air cannot be wholesome inside such a room. A place near the window

through which clean cool air is blowing in all the time is the proper location for breathing practices. Air is best when it is moving freely and naturally.

The fear of " cold draughts " is a medical superstition which is very much dying down even among them now, but is not yet dead. Cold air is the finest medicine for the sick; especially for those suffering from lung affections such as asthma. It makes the healthy healthier still. Readers should remember that any measure that does not have this benefit should not be acceptable in Nature–Cure.

We can become strong only by becoming *at home with Nature*, by obtaining the freedom of all the elements.

It is an assumed etiquete that we should tolerate others smoking in our presence. Should they not develop that etiquete not to smoke in the company of people who do not smoke? They have no right to pollute the air that is as much ours as it is theirs. If we do not gather courage enough to stop this in as many public places, like cinema halls, meeting places, buses, etc., then we will become smokers with the cigarettes between others lips. The injuriousness of tobacco smoke cannot be exaggerated

if we say that whether you are a smoker or not if you inhale it, you can be a victim of cancer of the lung or the throat or the mouth, or anywhere in the body for poisonous matter can accumulate in any weak part of the body dependent upon the constitution of the person. It will be a great service to the world, if each one of us keep on persistently advising our friends and others against smoking. A smoker becomes a slave to the cigarette, while he theoretically knows that it is bad for him. Remind him that he is no more the master, but a slave of a thing that he has paid for. Remind him that he purchases his own slavery. Ask him to be a man, standing upright, and not a slave to an insentient, lifeless being.

4. RESPIRATION

The respiratory tract includes the nose and naso-pharynx, the larynx, the trachea, the bronchi and the lungs. The whole tract is lined with mucous membrane. The mucous membrane contains certain cells known as goblet cells, secreting mucous and others with hair like processes called cilia, moving in only one direction (see Figure 6). The surface of the trachea and the bronchi are kept moist by the mucous secreted by the goblet cells. This condition serves to entangle particles of dirt that have entered the respiratory tract during breathing in (inspiration). The veaseless movements of the ciliated cells help in sweeping upwards, towards the mcuth, such particles as have been caught up by the mucous membrane. The ciliated cells act like a large number of minute brooms and manage to keep the respiratory tract clean. If, for any reason, the mass of foreign matter

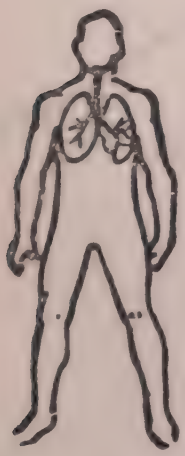


Figure 4.
The position of the lungs in the structure of the body.

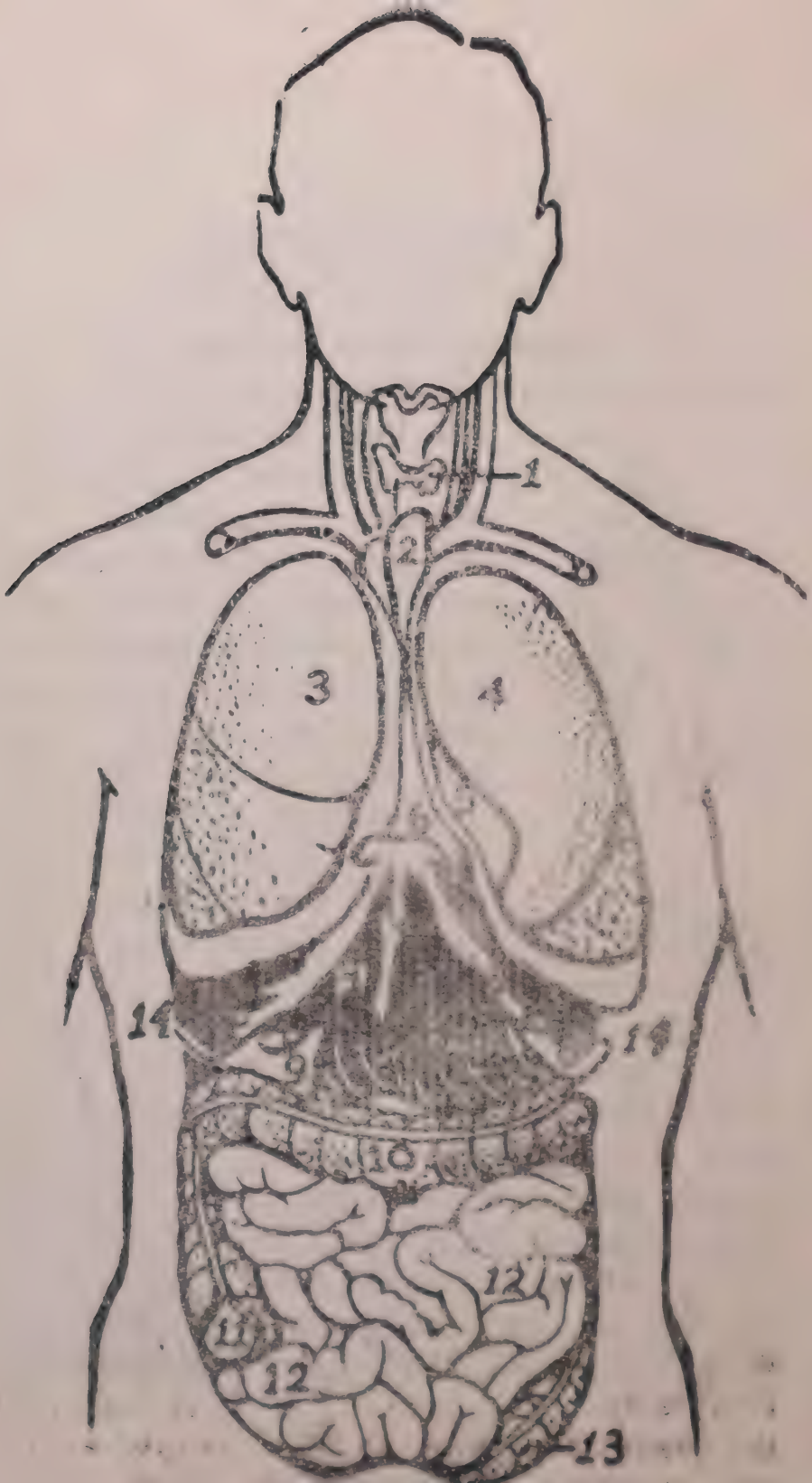

1
2
3
4
14
9
10
12
11
12
13

Figure 5. The Thorax

1. Thyroid
2. Thymus
3. Right lung
4. Left lung
5. Cardiac notch
6. Pericardium
7. Liver
8. Stomach
9. Gall bladder
10. Transverse colon
11. Ceacum
12. Small intestines
13. Sigmoid flexure
14. Rib cage

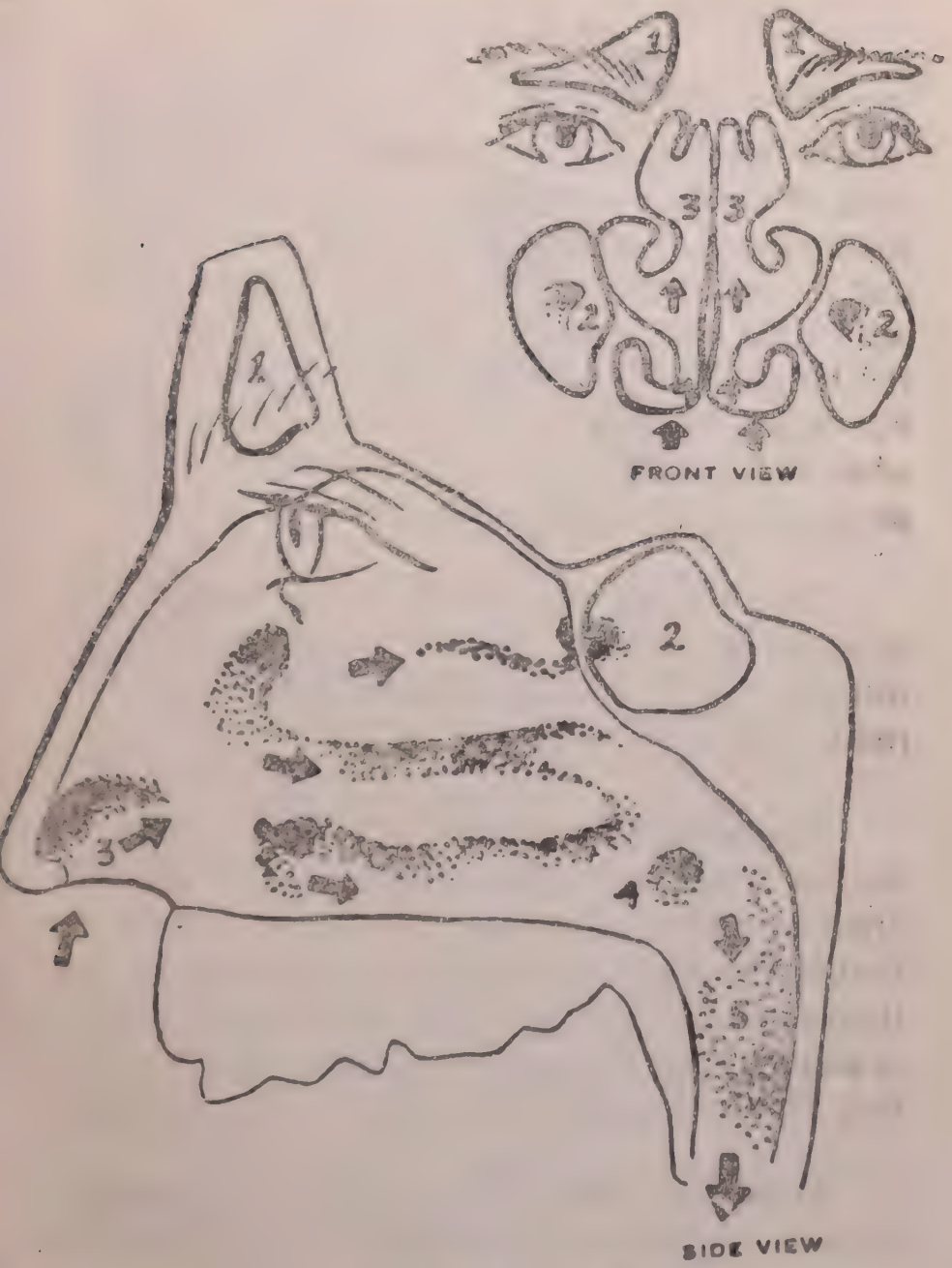
gaining entry into the tract be greater than could be managed by the ciliated cells, another process of cleansing is called for action, viz., sneezing and coughing. These are examples of the many protective reflex action which the body is capable of. The central nervous system provides that awareness to the various parts of the body and helps it to be free from foreign matter, be it dust, mucous or pus. Therefore an attempt to stop coughing of a person, whose bronchi are affected, is a serious mistake, and against the order of Nature.



Figure 6. The ciliated cells

The nose has two portals for the entry of air, separated by flexible dividing wall.

Figure 7 The Respiratory Tract in the Face



Top right corner: 1. Frontal sinus 2 Maxillary sinus
3. Nasal cavities 4. Nasal vestibule

Bottom left: 1. Frontal Sinus 2. Sphenoid sinus
3. Nasal vestibule of eustachian tube 4. Pharyngeal opening 5. Pharynx.

Its delicate lining is constantly active in removing impurities from the air that we breathe in. Smokers by-pass the safety mechanism provided in the nostrils by taking in the smoke laden with solid and gaseous impurities directly to the lungs. The hairs visible at the nostrils strain out larger particles. The city dweller is all too often made aware of the contamination of the air by the blackness of his nasal discharges.

The mucous membrane of the nose contains an exceptional number of capillaries. They diffuse a flow of warm blood which effectively reduce the chill of the air we breathe in.

Following the nasal cavity are pockets in the facial bones, called sinuses, which are also lined by ciliated mucous membrane similar to that in the nose. The sinus contributes to the fluid secretion, an increased discharge of which is experienced at times when the sinus is affected. This is a protective washing out process.

From the nose the respiratory passages turn downward at right angles to become the pharynx, which forms the throat that is visible through the open mouth. Because the pharynx has to play a dual role of (i) allowing food as well as (ii) air to enter downward, it is lined

with a tougher membrane more comparable to the skin than the lining of the nose. Delicate cilia can never survive the rough contact with food swallowed (generally not properly masticated food.)

The downward extension of the pharynx divides into the larynx for taking ahead air, and the oesophagus for taking down the food.

The entrance of the larynx is protected by a flap of cartilage, called the epiglottis (The thin leaf shaped flap of cartilage behind the tongue, which, during the act of swallowing, covers the opening leading into the larynx). While swallowing food the larynx rises and contracts so as to prevent the entrance of the matter, solid or liquid into the lungs.

The sound of the voice originates from the larynx. It has an important location in the front of the neck, well protected by a plate of armour the thyroid cartilage, commonly known as the "Adam's apple". This is more prominently sharp in males and is responsible for the distinct depth of the male voice.

Inside the pharynx the vocal cords are held in a stretched condition in the form of the English alphabet V with its apex on the top

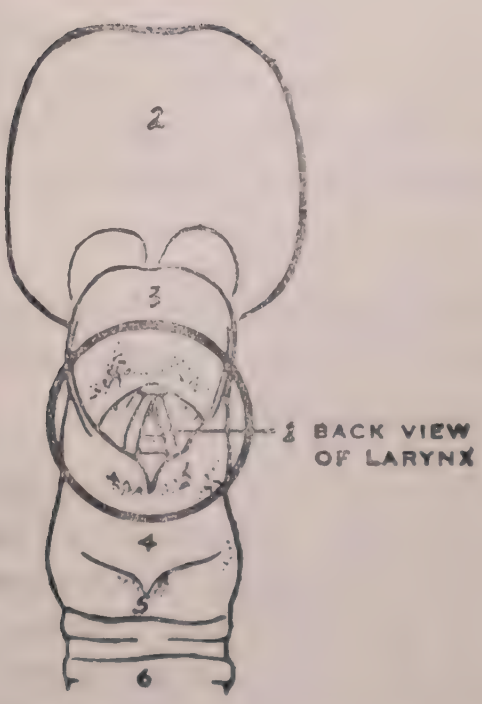


Figure 8

The Larynx

1. Vocal cords
2. Tongue
3. Epiglottis
4. Aryntenoid cartilage
5. Cricoid cartilage
6. Trachea

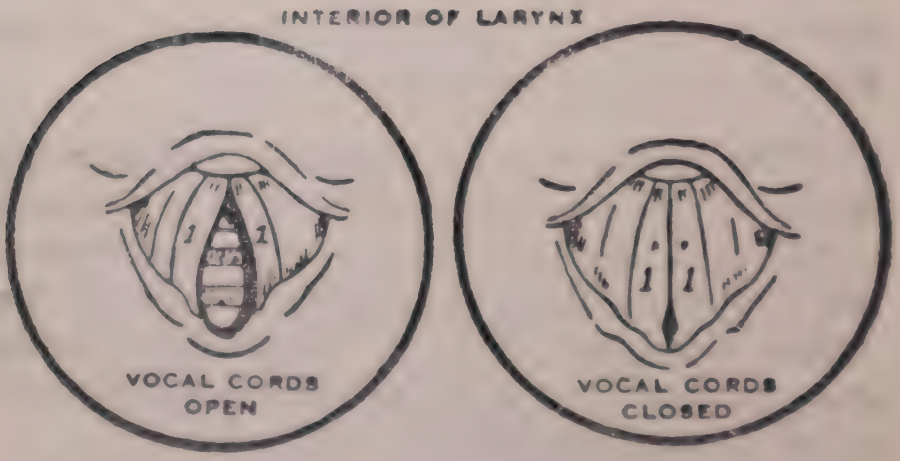


Figure 9. The Vocal Cords

The circled portion in the avove picture (Fig. 8) is shown greatly enlarged here.

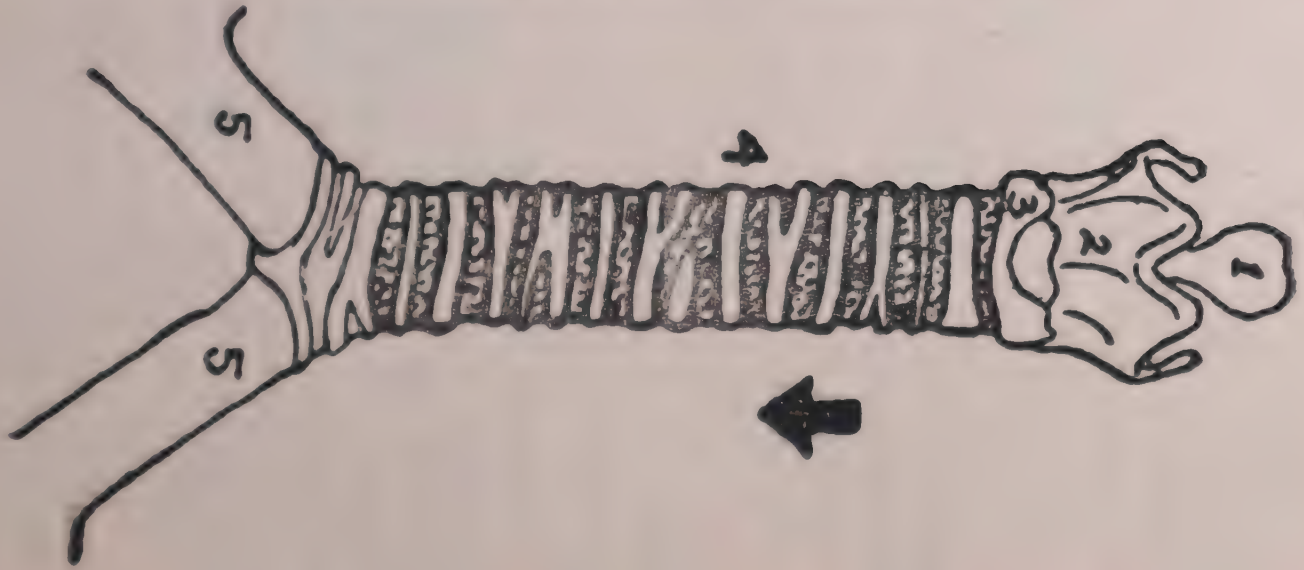
towards the front. They are tissues attached to the larynx with sharp folds. Sounds of varying intensity and pitch are produced by changes in the vibration and tautness of the vocal cords, while the air is forcefully expelled over them. The chambers of the mouth and the nose, and their cleanliness or otherwise account for the quality of the sound and make up for the difference between noise and music. The articulation of the lips and the tongue are largely responsible for the formation of pronunciation of words. Children who develop the habit of keeping the mouth open all the time get projecting teeth and later find it impossible or difficult to pronounce labial sounds, such as the letters p, b, m, etc.

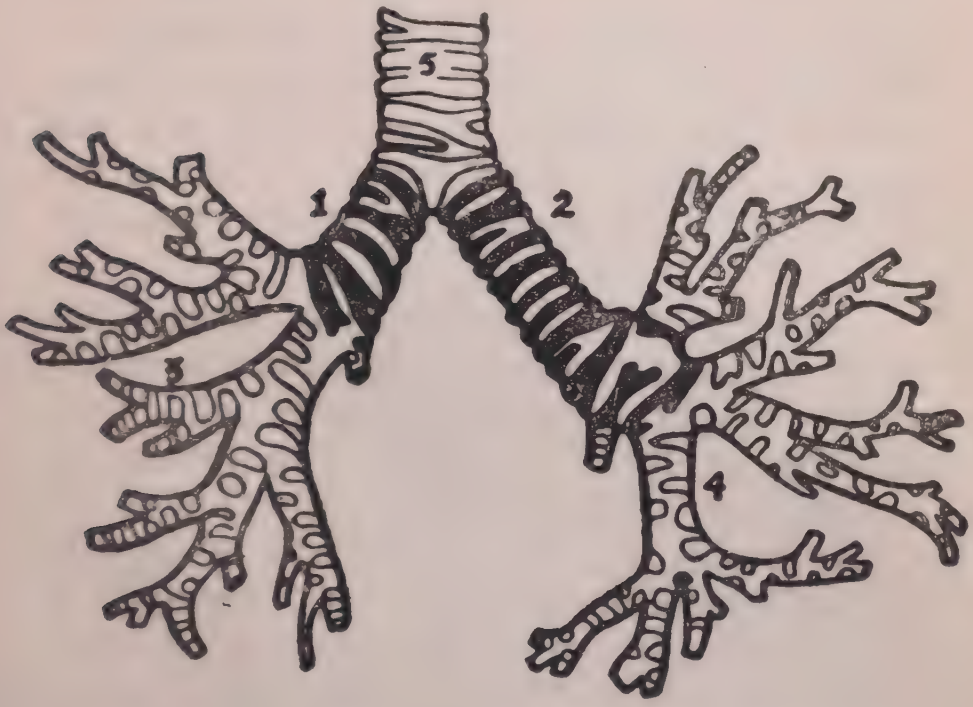
The trachea is formed into a cylindrical tube of about $4\frac{1}{2}$" in length by the telescoping of 16 to 20 crescentic cartilages and is fairly hard to touch. Its diameter is about $\frac{3}{4}$ to 1". Despite the fact that the wall of the trachea consists of firm cartilages, it can be compressed by strangulation. Wearing tight neck-ties do not constitute kindness to the trachea and the thyroid glands surrounding it. The absence of any discomfort is no excuse for the continuance of the mistake, because all such reactions are deadened with habit.

Figure 10 The Trachea front View

1. Epiglottis 2. Thyroid 3. Cricoid cartilage 4. Trachea
5 Primary Bronchi

The arrow indicates the direction of the passage to the lungs.





Figur 11 The Bronchi

1. Right primary bronchus
2. Left primary bronchus
3. Right secondary bronchi
4. Left secondary bronchi
5. Trachea

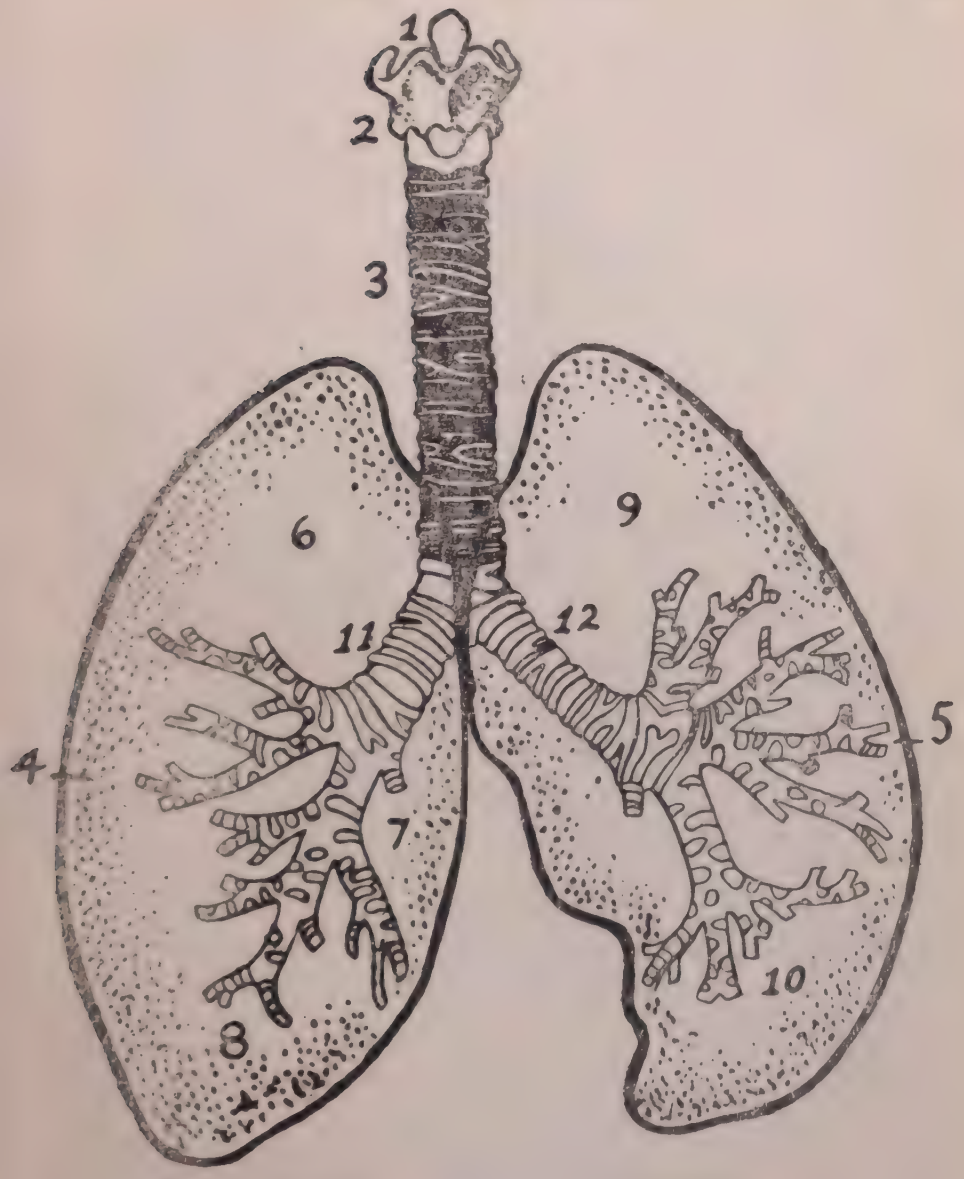
As the trachea enters the chest cavity, it divides into two branches, called the right primary and left primary bronchi, each one leading to one lung, after further branching off as secondary bronchi. This branching occurs at the level of the second breastbone or sternum. This location is of importance. The main artery of the body, the aorta, curves over it. The

oesophagus part of the food passage lies immediately behind it. Large lymph nodes that drain the lung encircle it. Naturally, enlargement of any of these structures, due to a diseased condition, may encroach on the space of the others and cause discomfort at this spot.

The right bronchus, which is shorter and wider than the one on the left, and composed of from six to eight cartilaginous semicircles, extends in three branches throughout the corresponding lobes of the right wing of the lung. The left bronchus which is longer and narrower than the right one, is composed of from nine to twelve cartilaginous semicircles, and enters two lobes of the left wing by two branches just under the aorta. These tubes are further subdivided into gradually diminishing, fine, reed-like ramifications, to the end of which are attached lung air cells, known as the alveoli.

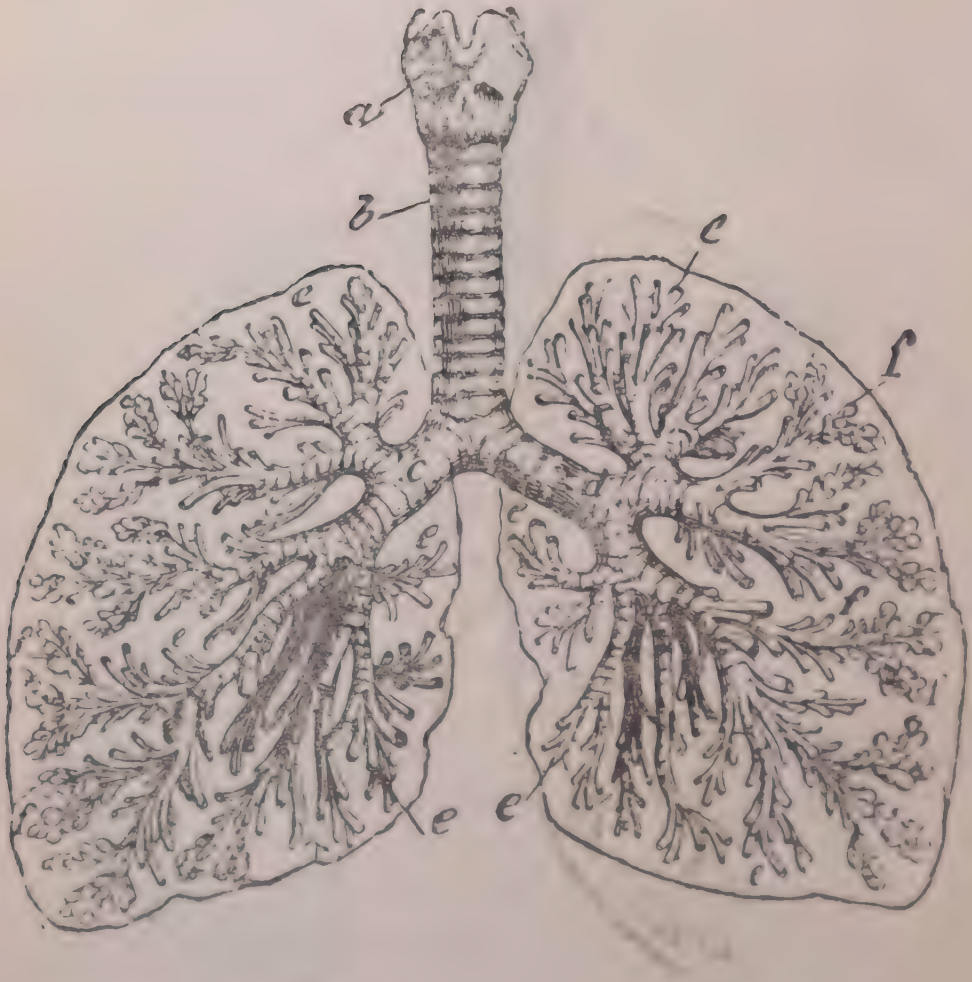
The tubes of the bronchi are elastic and are capable of expanding and extending. They vary in diameter with the expansion and release of the lungs during breathing. The small bronchi at the ends are of special interest. They are encircled by a muscle coat, heavier than elsewhere and capable of markedly narrowing the passages. Narrowing occurs when someone

Figure 12 The Lungs



1. Epiglottis
2. Larynx
3. Trachea
4. Right lung
5. Left lung
6. Upper lobe
7. Middle lobe
8. Lower lobe
9. Upper lobe
10. Lower lobe
11. Right bronchus
12. Left bronchus

Figure 13 The Lungs Showing The Alveoli



a. Adam's apple
b. Larynx
c. Left bronchus
d. Right bronchus
e. Branches of the bronchial tubes
f. Alveoli

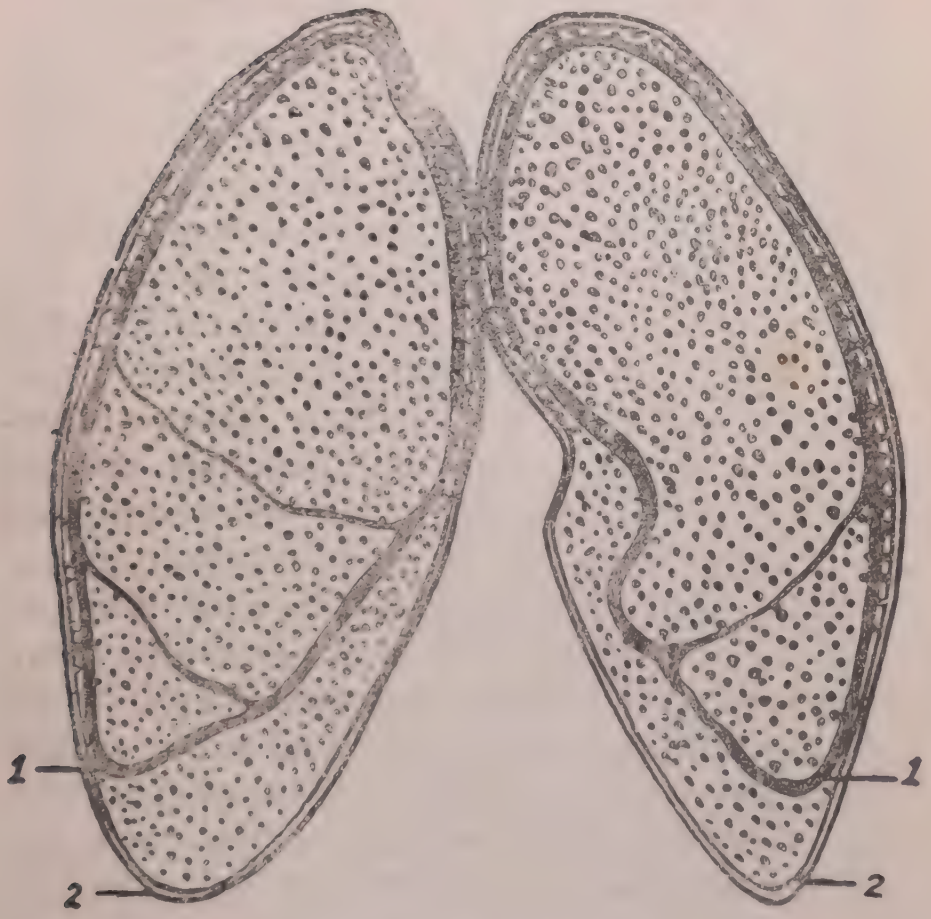
with asthma has an attack. We can hear the obstruction as a noisy prolonged period of expulsion of air past the narrowed small bronchi. This action is often mistaken to be one due to allergy. One can imagine that if the passage be wide the semi–solid phlegm could not easily be pushed up from the lung through the bronchi and then the trachea unless some special step is taken and that is the narrowing down of the passage, so that the morbid matter can be pushed up with a little lesser effort than normal. Drinking of water in aachamana doses, retaining each time for about 10 seconds in the mouth to allow it time to absorb the heat in the throat, the matter is enabled to dislodge and be pushed out more easily. Applications of a throat and chest cooling pack (a stimulating pack, if the stage of the patient's health demands), after an alternate hot and cold fomentation, preferably on the back, (because the asthmatic patient cannot lie on the back with comfort when he has an attack) greatly relieves the spasm, brings out plenty of phlegm with subsequent coughing and the patient gets rest, even sound sleep thereafter. Surgical interference in such cases, by the removal of a portion of the lung to improve the peristaltic action of the lung, to compensate for the weak bronchi, is in effect confirming a temporary

inflammation (itis) of the bronchus or bronchi The damage to the lung cannot be set right at all that way.

The right lobe of the lung lies on the right side and the left on the left side of the thorax or breast cavity. In the interspace between the two lobes, but a little towards the left is situated the heart. It is here reminded that the bronchus on the left is longer and narrower and branches off into three secondary bronchi on the left, while that on the right branches off only into two; and that is so to provide room for the heart.

The tiny branches of the bronchi are called bronchioles. The ultimate division into the air cells, which, budding from the bronchioles, may be likened to the leaves of the tree. The parallel between the air cells and the leaves is indeed a close one. It is through its leaves that a tree breathes, and it is through these air cells that an animal takes up oxygen and gives off carbon-di oxide. As the branches of the pulmonary (the breathing organ, the lung-adjective of) system get smaller and smaller their walls become correspondingly simpler and thinner, until finally they are so delicate as to consist only of a single layer flattened cells held together by a minimum of connective tissues.

Figure 14 The Lobes Of The Lungs & The Pleura



1. Lower margin of the lungs
2. Lower margin of the pleura

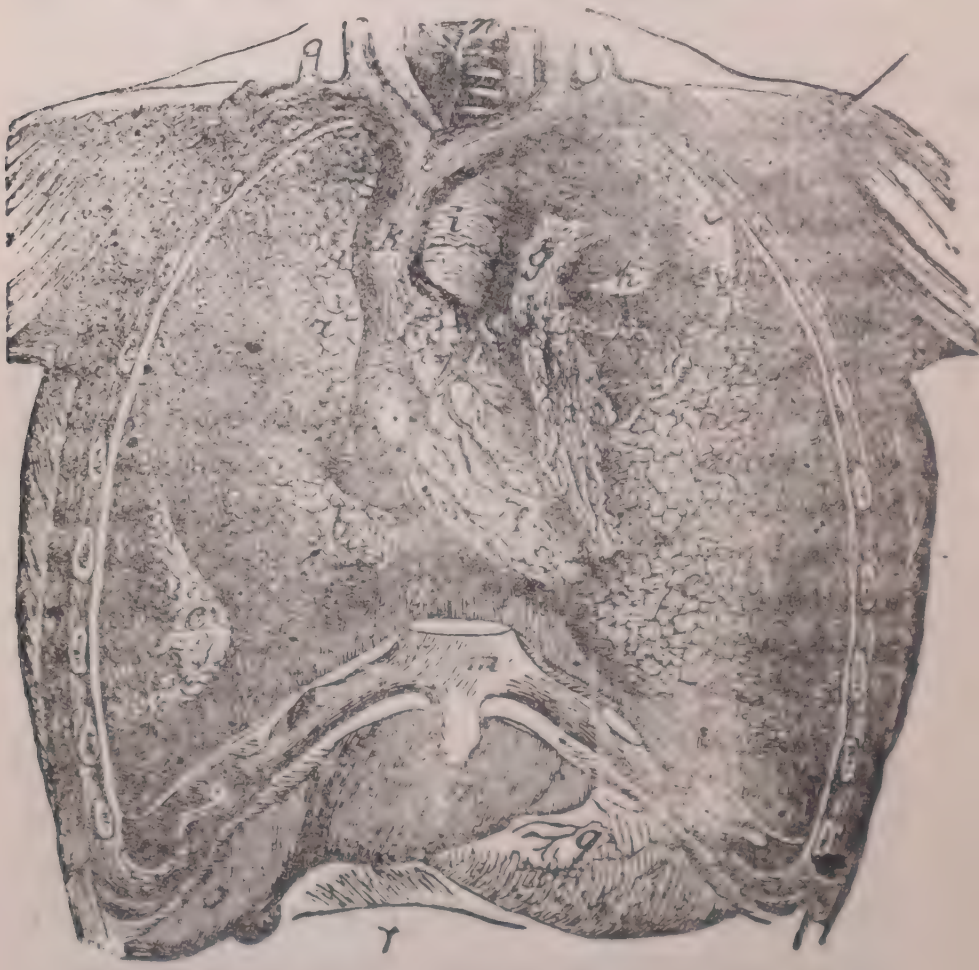
The lungs are voluminous organs, but much lighter than the liver. Each lung, divided into several lobes, is surrounded by a double layer of smooth shining membrane, the pleura.

That part of the membrane which is adherent to the lung is known as the visceral pleura, and that part which lines the inner surface of the chest wall as the parietal pleura. The smoothness of these two linings allows the lungs to slip over the inner surface of the chest during breathing, with the minimum friction. In an inflammed condition of the pleura, this movement no more remains smooth. The smoothness of the surface is lost (temporarily, of course–though the medicos believe otherwise), and the movement becomes difficult and painful and naturally the patient complains of distress whenever he has to take a little deep breath. Nature manages to reduce this discomfort and serious damage to the pleura by forming a secretion of a fluid between the two layers of the pleura. Nature expects of every patient to realise this his path (the course of living) has been wrong and demands immediate correction of those wrong ways that made him sick. Therefore continuation of the same wrong ways of living is liable to turn this fluid purulent (pertaining to or resembling pus), and such a state is called empyema (collection of pus in a cavity, hollow organ or space). Allopaths evacuate the pus after making an incision into the chest wall. (This operation had to be performed on King George V during his last illness).

" The lungs of a newborn baby are pink, but most adults have taken in enough dust so that their lungs are coloured grey with black streaks." Lungs resemble sponge only superficially. They can be squeezed much as sponge can be done, because both have many air filled pockets, but the structure of the lungs are more orderly than that of sponge. All the innumerable branches (bronchioles) of the bronchial tree lead into air sacs known as the alveoli (plural of alveolus). An alveolus is the ultimate unit of the respiratory system and is the spot where gas interchanges between air and blood. The alveoli are fine microscopic cavities, quite imperceptible to the naked eye.

Each lung has a rounded apex at the bottom at the junction of the outer side with the base which is widened over the diaphragm, which is a wall separating the abdomen from the chest. The right lung is divided into three lobes, while the left has only two. Here the reader is reminded of the construction of the two bronchi, the right and the left. All these are to provide enough space for the heart, which is a little to the left of the thorax.

Figure 15 The Organs of The Chest (laid open anteriorly)



a. Right upper lobe
b. Right middle lobe
c. Right lower lobe
d. Left upper lobe
e. Left lower lobe
f. Coronary artery
g. Pulmonary artery
h. Pulmonary veins
i. Aorta
k. Superior vena cava
l. Diaphragm
m. Appendix of the breastbone
n. Trachea
o. Right lobe of the liver
p. Left lobe of liver
q. Stomach
r. Transverse colon

These lobes are further divided into little clustering cells, filled with countless nerve, lymph and capillary vessels, joined to each other by connective tissue.

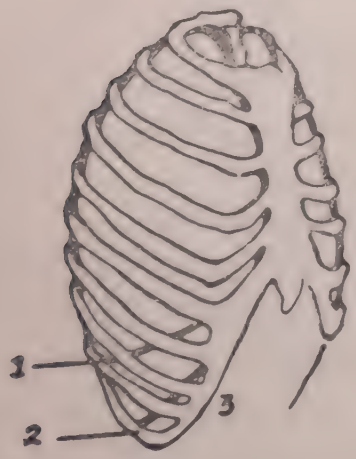


Figure 16 The chest bones
1. Lower margin of the lungs
2. Lower margin of the pleura
3. Lower rib margin

The right and the left pleura, each of which is covering that particular side of the lung, compose the pleural cavity. The space between the right and left pleura is called the media–stinum. This space contains the heart, the trachea, the oesophagus and the large blood and nerve vessels.

The lungs encased in the pleura are suspended in an air–tight box, the thoracic cavity. The walls of this cavity, consisting of the ribs, the sternum (the breastbone; a narrow, flat bone shaped like a dagger with a handle, blade and tip; the first seven cartilages are attached to it on either side; it articulates with the clavicles or the collar bones above), the vertibral

column and the diaphragm are movable. The shape, size and chest determine the freedom of expansion of the lungs while breathing in. It is needless to say that the lungs will have to contract with the diminution of space in the chest cavity. Similarly it does not leave any space in the thoracic cavity whenever there is an increase of space by the expansion of the chest: Such an expansion occurs by two mechanical actions, viz., an upward movement of the ribs and a downward movement of the diaphragm. Since the ribs are placed obliquely, sloping downwards and forwards from the vertibral column to the sternum, it is obvious that when they are raised the capacity of the chest will be increased. This is achieved by the muscles of the chest wall, and this process is synchronised with the downward pull of the diaphragm. This forcible descent of the dome of the diaphragm increases the pressure within the abdomen, with the result that the abdominal wall is pushed out–forward. All these co–ordinated movements of the chest and diaphragm, together with the drawing of air into the lungs, constitute the act of inspiration.

A complete reversal of the above processes will constitute expiration. The diminution of the capacity of the chest may be either active

or passive. In ordinary quiet breathing no muscular action is needed to bring back the chest wall to its resting position. The natural elasticity of the tissues and the weight of the chest are sufficient for this purpose. But during forced, or laboured, breathing additional help is afforded to expiration by the muscules of the abdominal wall. By contracting, these muscles increase the intra-abdominal pressure and force back the diaphragm towards the chest. The part played in breathing by the chest wall and the diaphragm differs in different individuals. The general observation is that women are more dependent on the chest breathing while men make relatively greater use of the diaphragm. This is so in the case of men because of lack of instruction in childhood, both in eating as well as breathing habits and they have developed a rigid chest. Women desiring a narrow waist, wear their clothes tight at the waist and have naturally to utilise the elasticity of the chest for breathing. This is only a seeming benefit. Restriction on any side is wrong. This has been explained earlier. A narrow waist should be the gift of abstemiousness as the great Tamil poetess Avvayaar has said, "உண்டி சுறுங்குதல் பெண்டிற்கழகு" (Abstemious eating bestows beauty on the belly.)

In normal breathing, all the movements of the lungs are entirely passive, imparted to them by the movements of the chest wall. The respiratory tract contains a small amount of muscular tissue. The bronchi and the bronchioles are surrounded by a thin layer of muscle, which when it contracts diminishes their lumen (the interior of a tubular structure).

During occasions of prolonged muscular exercise, such as is entailed in running and other manual labour, rapid and successive contraction of various muscles happen, causing an increased expenditure of oxygen and, when it has been utilised, plenty of carbon-di-oxide has also to be thrown out. This necessitates more rapid breathing--both in and out. This increased frequency of respiration is an automatic action brought about through the central nervous sysem. There is respiratory centre in the brain, in that part which is continuous with the spinal cord. The increased amount of carbon-di-oxide circulating in the blood acts upon this sensitive centre and causes a quickening and deepening of breathing. This quickening of the breathing can be said to be because of a chemical stimulus on the nervous system and that in turn brings about a fall in the carbon-di-oxide content of the blood. If the cause for

this stimulus ceases to exist, obviously the chemical stimulus also is removed and breathing again becomes normal. If the exertion be so severe and so prolonged as to cause such a deep and quick breathing as one's muscles and nervous system could not cope up with dyspnoea or difficult or laboured breathing or breathlessness will result.

The reason for mountain climbers and airmen having to breathe more deeply and more rapidly is that at high altitudes the oxygen content of the atmosphere is comparatively lower than at the plains. It must be remembered that even as the rate of breathing increases to be able to supply the blood stream with more oxygen and also to scavenge it of the carbon-di-oxide that it carries back from the tissues, the rate of pumping blood by the heart has also to increase in tune with the lungs. People with a weak heart find difficulty therefore at high altitudes and dyspnoea is caused. Some people who might just be able to manage at such high levels cannot accept any kind of exercise—physical and physiological. Digestion of food is one of the latter type and vomitting will be the result.

The lungs of the baby just before birth are in a collapsed state. Before normal breathing can start, it is necessary to expand the lungs. The lungs are provided with elastic fibres in the alveoli and bronchi. But they are intended for normal functioning of the lungs. This extra-ordinary necessity has therefore to be achieved with their help, aided by the respiratory muscles, by a vigorous cry that is always seen when the child comes out. In rare cases this does not happen and the midwife induces this cry by slapping the child on the back. It should not be mistaken for an unkind act, but a necessary one to see that the child starts normal breathing. Once the child cries, the lungs expand and never again collapses in life. Even by respiration, with all the extra care to see that it is completely emptied, there is some residual air left behind. This aspect will be explained in detail in the next chapter.

The composition of the atmospheric air is more or less as follows:

Oxygen	—10·96	volumes per cent
Nitrogen	—79·00	" " "
Carbon-di-oxide	— 0·04	" " "

It also contains a varying amount of water vapour; though rarely saturated with it. The temperature of the air, of course depends on the season and region of the earth.

In the respired (breathedout) air there is a decrease of the oxygen content with a corresponding increase of carbon-di-oxide. Generally, the composition of the respired air in the case of man is as follows:

Oxygen	—16·6	volumes per cent.
Nitrogen	—79·00	„ „ „
Carbon-di-oxide	— 4·4	„ „ „

It is also nearly saturated with water vapour and on a cold day it can be observed to be condensing in a cloud of steam with every breathing out. Its temperature is about 36° C (96.8° F), being very slightly affected by the external air. The volume of the expired air is about 1/50 less than inspired air.

Therefore, out of the oxygen inhaled, 4.36 volumes per cent, is used up, and the same returns in the exhaled air in the form of carbon -di-oxide. From these two data, viz., the percentage of carbon-di-oxide in the expired

air and the percentage of oxygen in the inspired air, we can arrive at the respiratory quotient thus :—

$$\frac{CO_2 \text{ expired}}{O_2 \text{ inspired}} = \text{respiratory quotient}$$

The respiratory quotient, according to the data given above will be $\frac{4.36}{4.36} = 1$. But this respiratory quotient is not the same figure with all kinds of diet. On a pure diet of fat, the respiratory quotient is only 0.71, because a part of the oxygen inspired is utilised in oxidising the fat:

$$C_3H_5(C_{18}H_{35}O_2) + 80\ O_2 \rightarrow 57\ CO_2 + 52H_2O$$

Therefore out of the 80 molecules of oxygen only 57 are expired as carbon-di-oxide and the rest of the 23 molecules go to oxidise the fat, being thrown out as water vapour. Similarly with a diet which contains more of proteins the respiratory quotient is again less than that with a herbivorous diet, because the oxygen is utilised in this case to oxidise the nitrogenous matter of the proteins, excreting a great part of it as urea ($CO(NH_2)_2$). Thus it follows that in the case of vegetarians, the oxygen inspired is fully utilised for the actual purpose of breathing, i.e., interchange of gases on the

blood arriving in the lungs, and hence ensures Vital Economy,

Liquids are capable of carrying gases in solution. The amount of gas that the liquid is capable of taking in solution varies directly with the pressure of the gas. At a particular pressure, whether the gas be one (pure) or a mixture of gases, the total quantity dissolved will be the same. Therefore, if it happens that there is a mixture of gases at a particular pressure, it will mean that one of the gases exerts only a part of that pressure proportional to the percentage of its own content in the mixture of gases.

Similarly, when there is a reduction of pressure, there will be a release of gases dissolved in the liquid.

The liquid will also give off the gas that it held in solution, if there is a rise in temperature.

The interchange of gases both in the lungs and throughout the body is governed by the above said properties of liquids and gases.

There is just one difference in that the oxygen or carbon-di oxide is carried by the blood not in the form of a simple soultion, but

by the formation of a loose, easily dissociable form of chemical combination wtih red blood cells called haemoglobin which contains iron because of which it has the colour and capacity of being a carrier of oxygen. But the haemoglobin does not take in all the gases that are presented to it in the lungs along with the oxygen in the atmospheric air.

This therefore drives us to the conclusion that the pressure exerted by oxygen alone in the atomospheric air is only (12.6 to 13.5 percent) about 1/7 of the atmospheric pressure. Thus at a reduced pressure, haemoglobin leaves off carbon-di-oxide and takes in oxygen.

It was said that haemoglobin forms a loose, easily dissociable chemical combination with oxygen or carbon-di oxide. But it can have a more stable combination with carbon monoxide which is poisonous and thus cause a permanent loss of the greatly valued red blood cells capable of carrying oxygen. Carbon monoxide has neither colour nor smell and inhaled continuously for some time will cause unconsciousness, asphyxiation and death. Flames at a high temperature emit this dangerous gas and should be avoided. Extreme care should be exercised in seeing to it that there is proper ventilation

when using such fires (coal, petroleum or butane gas, etc.,) for cooking purposes. Baking the roti or chapati over such fires is equally bad. Raising them to a puff after an initial baking over the dry pan (tava) should preferably be done on a low fire with charcoal.

If the oxygen content of the atmosphere is less than 3 percent there will be a rapid progress to death with the stages of asphyxia and convulsions. If the atmosphere happens to contain 6 percent or more of carbon-di-oxide death will of course occur but with different stages, viz., hyperpnoea (panting) and gradually narcotised effect *without* covulsions.

Gases such as Nitrogen, Hydrogen and Methane (CH_4) are termed indifferent gases, and any of the latter two gases can replace Nitrogen in the atmosphere and provided the oxygen content remains the same, there will be no danger.

Hydrocyanic acid, sulpheretted hydrogen (H_2S) and phosheretted hydrogen (PH_3), arseniuretted hydrogen, etc. are also poisonous gases, besides carbon monoxide.

Ammonia, chlorine, sulphur dioxide, nitric oxide and other gases are irritating to

such an extent that they produce a spasm in the glottis. But people living in cities with the so-called protected water supply are compelled to consume a daily dosage of chlorine in solution in the drinking water. It is added to kill the invisible germ immediately and *the consumer gradually*.

"An adult man gives off about 0.6 cubic foot of carbon-di-oxide every hour." The normal amount of carbon-di-oxide in the atmosphere is 0.04 percent. Hence he raises, in an atmosphere of 1000 cubic feet, the percentage of carbon-di-oxide to 0.1 percent. Therefore, it further follows that if 60 men have to share that volume of atmosphere for an hour, the air will become so dangerous as to cause a mass death by carbon-di-oxide poisoning, as explained earlier.

"In order that the air may be easily renewed without giving rise to excessive draught, a certain amount of cubic space must be allotted to each man. Each adult should have in a room 1,000 cubic feet of space, and be supplied every hour with 2,000 to 3,000 cubic feet of air."

At the height of inspiration the blood contained in the lungs is about 8.3 percent of

the whole blood content of the body. During expiration it is reduced to something between 6·5 and 5·5 percent. There is a mutual relation between the respiratory variations and the systemic blood pressure. With each inspiration the blood pressure rises and with each expiration it falls. Each operation of rise or fall continues for some time before the next one follows. To be clear, the two are not simultaneous, but the blood pressure being the result of the respiration, it follows breathing and continues till a change occurs. Thus the highest point of pressure occurs at the beginning of the expiration and the lowest at the beginning of the inspiration. The heart beats during the fall that accompanies the expiration.

Any fluid will flow from a point of higher pressure to one of lower pressure; and the blood is no exception to this rule. But the fact remains that both the arteries and veins are not affected in the same manner. The reason for the difference is that the arteries have comparatively thicker and non-distensible walls, and the pressure outside it has little effect on it. At the end of each expiration the heart and the large blood vessels are subjected to a negative pressure (of -6 mm. Hg.), while outside the thorax all the blood vessels are

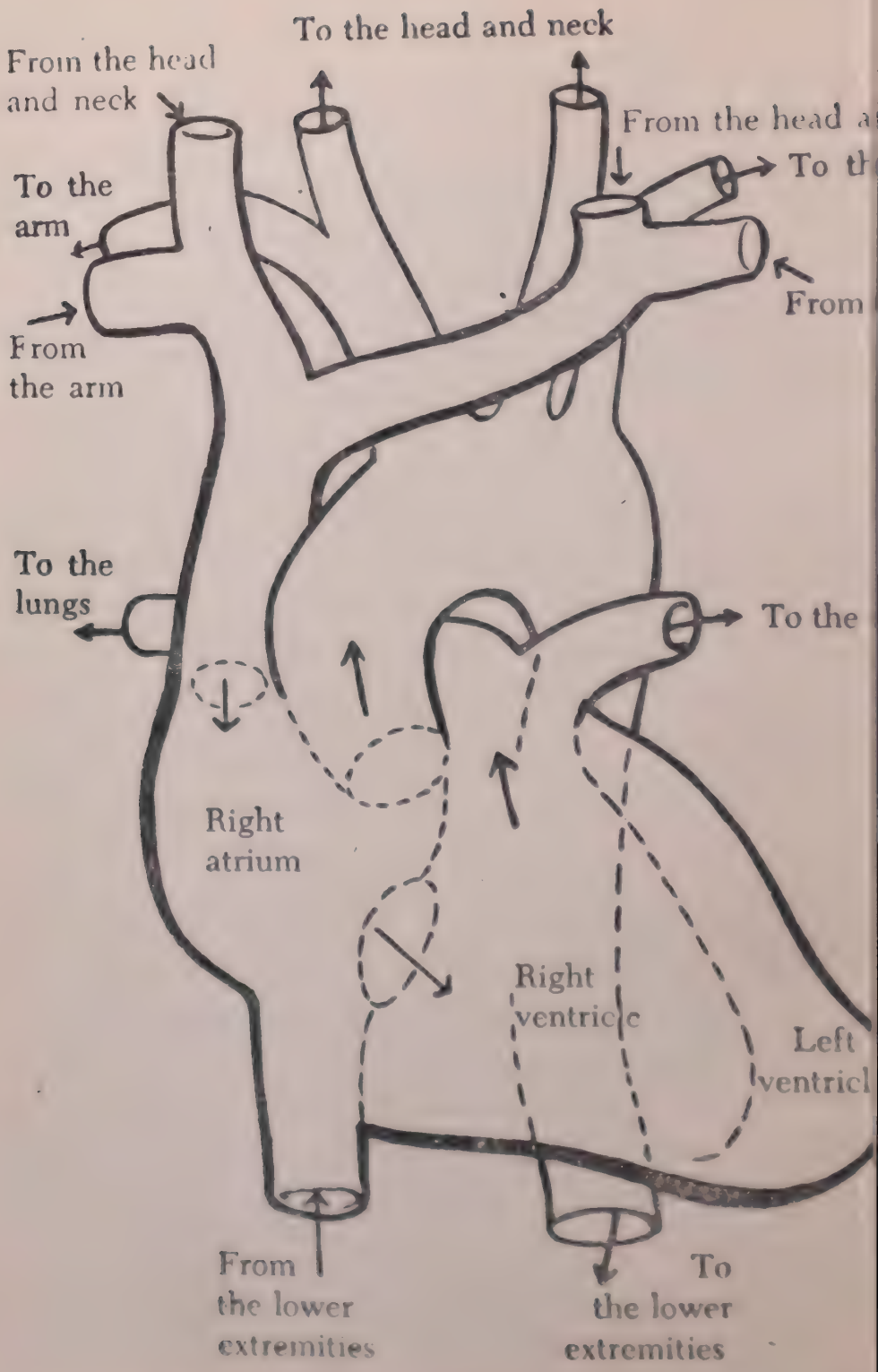


Figure 17 The Heart—Main Blood Vessels From And To The Heart

exposed to a positive pressure because of their own carbon-di-oxide content as well as by the elasticity of the tissues in the neck and by the contractions of the diaphragm and the abdominal muscles. Obviously, the flow of blood through the veins into the heart is greatly influenced, without affecting the outflow of blood in the arteries, because of reasons explained earlier. This process therefore starts at the end of each expiration and attains the peak by the time the inspiration ends, resulting in an increase in the arterial blood pressure, thereby enabling an easy outflow from the heart. From this it follows that the heart is able to suck in large quantities of venal blood and push out in the arteries, greatly enabled by the depth of expiration, which is automatically followed by a better inspiration. This equally influences a better exchange of gases in the lungs, because the deeper the inspiration the greater the surface area of the alveoli (about 2,000 square feet), whereby every blood corpuscle is being practically exposed on two sides.

This gives a scientific reason for the preference of attention towards better Rechaka, which will automatically be followed by a better Pooraka.

5. PRACTICE OF BREATHING

The fitness of the lungs to serve as organs of breathing depends upon their elasticity–a quality by which they shrink and swell alternately. Healthy lungs shrink to a very small size, squeezing the air-cells (alveoli) so as to empty them of nearly all the air, and then expand to a large size so as to create ample vacant spaces in those cells for the incoming air to occupy.

The elasticity of the lungs depends upon the purity of the tissues of which the organs are made and of the nerves by which power flows into them. Of course the purity of the body substance depends on these things: Food includes fasting (because Ether is a food that can be had only while fasting and Ether is one of the Five–fold food of all beings)and exercise. Wrong or excessive food and insufficient exercise will thus make the lungs inelastic.

It was explained in the previous chapter that the space available for the lungs can be increased to a maximum by the ribs raising up and forward to provide more space in the thorax and the diaphragm (the dome shaped

muscular partition between the thorax above and the abdomen below) depressing down while the abdomen bulges out. If the wall of the abdomen had been stretched to its limit already by the person having developed a protruberant belly, it may not be further possible for it to bulge out. If the abdomen is not free from accumulations, the diaphragm cannot find space for it to go down.

It was also told in that chapter that these involuntary functions of the body are constantly being controlled and regulated by the central nervous system. Therefore, if power and as well as messages do not flow freely and in time, the functions of the lungs can only be under strain. The primary need therefore is of restoration of cleanliness and thereby the elasticity of the vital organs at least to some extent, so that the practice of Prayanayama becomes possible. A few years of dietetic righteousness, with a little more of general exercise, will so purify the organs that the desired effect can be had. It may be that in a few cases the desired extent of relief is had even in the course of some months. In fact it depends upon the Vital Reserve, the extent of reform in eating, working and other habits, the quality of the available food, which includes all

the Five–fold food, and above all the strength of conviction, fearlessness and mental abandon to accept the consequences as they come and not being the worrying type, constantly putting the the question, " Why so late?".

Even with a little improvement in the general health, one can begin the lesson in practice, knowing full well that no rapid progress is to be expected *without refining the lung substance by the practice of dietetic righteousness.*

Lowering of the health level occurred because of accumulation of morbid matter and the organs of elimination having been unequal to such a heavy task. Naturally, this indicates that most of the organs of elimination, some of them vital ones, like the lungs and the kidneys, have been made to work beyond their normal capacity. Even a strong man needs rest after hard work. When one is either constitutionally weak or weakened due to overwork, rest is all the more necessary.

Rest can be provided to the system in so far as the voluntary functions are concerned. But in respect of organs that look after the involuntary functions, functions with the failure of which even life will fail, allowing rest

is not completely in our hands. That means to say that it is somewhat in our hands too.

We can manage to see that the work normally shoved upon them is greatly reduced through fasting or by remaining on very low diet of highly positive, alkaline, bulky and cleansing foods, instead of highly nutritious, concentrated, sticky like the binder's paste, and acid forming ones. Of course such earthy[4] matter alone does not constitute our food: there are the finer elements, of which Ether is availed of while fasting. The rest of them, Air, Light and Water should also be had in the purest form possible. In city life it is of course rare that people get any of them pure and in plenty. Air is polluted by fumes from mills and factories and running automobiles and the etiquetless smokers of tobacco in any form. Working in huge buildings with low ceilings and sealed windows for efficient air-conditioning, both in summer and winter, people deny themselves the bountiful sun's rays. Their intention may be good; but the addition of chlorine and other poisonous additives to drinking water, in the

4. Of the Five-fold and therapeutics Earth is often taken as mud by ill-educated naturopaths. Earth, if we are to take it as mud, where is the place for Diet in the Fivefold food and Medicine? Food from plants grown on the earth is the real earthy food. Earth or mud as such is only a good vehicle for water.

name of "protected water supply",[5] public health is endangered. If these conditions are unavoidable, sufficient measures of compensation should be taken by having an early morning walk, drinking clean spring or river water; of course rain water, whenever available, is the best that could be thought of, if it is not polluted by factory chimneys.

Through every one of these measures, the burden of elimination of wastes can be greatly reduced. That will, compared with the hard work that was put on the system, mean resting the organs.

However, there are certain organs, vital as they are, that cannot afford (to go on leave) to rest, while we are alive. They are the heart and the lungs. It is said that the heart gets rest during every diastole (the relaxation period of the cardiac cycle). But the lungs....., could we not rest it too? That is the problem for which a solution is there.

---

5. If you live in a city with "protected" (polluted) water supply and no good alternative exists, collect water from the shower at a good height (nearly 6 feet), so that the water gets a chance of being aerated to dissipate chlorine to some extent. It can further be improved by making 'Hamsodakam' by placing water in a wide mouthed vessel exposed to sun by day and the chill air of the night, preferrably with a thin clean white cloth or a banana leaf for covering, to protect the water from getting dust, etc.

One Pranayama consists of three steps or processes (in the traditional style) known as Pooraka, Kumbhaka and Rechaka, meaning respectively inspiration, retention and expiration. All these three steps could be considered exercise. Obviously, pleasure seekers and weak people rarely exhibit any inclination to take up the practice of Pranayama. Even people of the first group, though looking healthy, are really lacking that high-level-health which makes work a pleasant one.

The purpose of exercise should be to provide to parts or organs that are normally inactive or that do not have a need to function in all angles. But exercise should not be ones that tire out the person. It is essential that there be a harmonious combination of both exercise and relaxation.

In the Pranayama, as set out above, all the three steps are ones that provide work or exercise. There is none that provides a rest. Yogis claim that Kumbhaka is a period of rest and relaxation. It may be so theoretically; but not to those who have not attained that advanced stage of perfection. We are more concerned with the beginner and those who

take to this as an additional aid to the restoration of health. For an example, sitting absolutely idle or remaining in bed all the time when there is a little ability to move about, is more taxing than doing a little exercise. Still more difficult is the keeping of the mind without thought while one is awake. Retaining the breath inside and forgetting about it for a length of time is easy for one who has attained that stage in which there is effort in doing so, not for him for whom the mind is all the time conscious of what he is doing.

It is argued that the interchange of gases occur in the lungs almost within an instant and the remaining period of the kumbhaka can therefore be considered a period of relaxation. May be it is so, but it cannot be denied that it is coupled with a control, and therefore a strain. Even so, if we take it as a period of rest, the steps are one step of work, one step of rest, and one step of work, following which will come the first step of work of the second Pranayama. Thus, at the junction of two Pranayamas there is also a coupling of two steps of work and the step of rest remains one and this spoils the rhythm. When our aim is to establish a rhythm of work and relaxation, this falls short of it and remains an exercise, unacceptable to the health-seeker.

Therefore, to break this monatony of two steps of work coupling together at this junction of two Pranayamas, viz., Rechaka and Pooraka, if we introduce a gap of inaction between these two, a cycle of Pranayama is to have one step of work followed by one step of relaxation, another step of work, followed by another step of relaxation. When the steps are even in number and work and relaxation alternating, the rhythm will be kept up even at the junction of two Pranayamas.

This new step that has been introduced, though it be one of relaxation, because it is a period of retention (of air outside the lungs), is given the name Baahya Kumbhaka. Now that we have two steps of Kumbhaka, though interrupted by Pooraka, it is necessary to distinguish between the two. Therefore we call the retention of air outside as Baahya Kumbhaka and the retention of air inside the lungs as Aantarik Kumbhaka.

We know that breathing is a vital process, aimed at feasting on the second of the Five-fold Food. This food is definitely far more superior and finer when compared to the earthy food (through plants, of course). We

stress the need to see that the stomach should be clean before we eat. Similarly, it is even more necessary to see that the lungs are cleaned before we partake of that superior food, Air. *We therefore give the step, Rechaka,* (expiration), *the foremost place in the course of a Pranayama.*

The amount of air that can be inspired depends upon (i) the elasticity of the lungs and (ii) the vacant space created by the expiration. This also adds to the importance of the Rechaka with emphasis on perfection.

In normal breathing, an act going on without our being conscious of it, we breathe in or out 250 to 300 c c. of air. This amount is known as the *tidal air.* By means of a forcible expiration, the contraction of respiratory muscles will drive out about 1,500 c.c. or even a little more. This is known as *supplemental air.* When next we inhale, about the same amount of air can be breathed in, of course conscious of the fact and with effort to take in the maximum possible. This is called the *complimental air.* But however much one tries to expire, one cannot remove all the air from the lungs; if it is done, the result will be a vaccuum inside the lungs; in such a case the lungs will

collapse to the pre-natal stage and it may prove dangerous. The amount of air left behind inside the lungs is called the *residual air*. This is said to be nearly 2,000 c.c.

The supplemental or the complimental air, individually is less than the residual air. The data provided above are from medical text-books, whose authors are unaware of Prana-yama. It is our belief that by constant and persistent practice, the residual air can be made less and the supplemental and complimental air increased. We regret we have not the facilities for carrying out research on the subject on the lines we would have them; we hope we may be able to do so in the not distant future.[6]

In ordinary breathing the residual air is so excessive that the fresh air rarely enters the alveoli (air-cells), but stops farther away, and the exchange of gases has to take place—if at all it can—through the medium of the stagnant residual air. The fresh air can be brought into

6. Sri L. Ganesa Sarma was requested by Sri Kurur Nilakantan Nambudripad of Trichur to prepare a scheme for starting a National Institute For Research In Natural Therapeutics. He has prepared one. The friend appreciated the scheme, but by the time it was ready, due to certain political changes, the possibility of establishing a research institute faded. It is now open for any fresh offer and The Indian Institute Of Natural Therapeutics will ever be prepared to provide all possible help in its power.

the cells only by reducing the residual air. If this is not done, degenerative changes in the very substance of the lungs may begin. This should therefore be avoided by keeping up the practice of Pranayama as a regular daily routine on more than one occasion in the day. People who do it as a religious observance are expected to do it thrice in a day, during Praata Sandhyaa, Maadhyaanhikam and Saayam Sandhya. But there are people who keep up this regularity of just repeating the mantras and doing the gestures, but having no knowledge or pracrice of the art of Pranayama. Practice must help to perfect the normal breathing as well.

Reciting the mantra is thought to be unimportant by most people. We would take it for a hasty judgement. Leaving aside the intrinsic merits of the mantra, which are unquestionable, we cannot be blind to the fact that the recitation of the mantra has the welcome effect of detaching the mind from the effort in doing the different steps of the Pranayama.

The mantra is given as an appendix and those interested in it can look into it and *learn it through a guru*, if possible. It should be split

into four parts to provide for the four steps of a Pranayama, viz., Rechaka, Baahya Kumbhaka, Pooraka and Aantarik Kumbhaka. Such a division is also indicated therein.

We referred to rhythm in breathing. Therein we mentioned the need to alternate work and rest. There is yet one more rhythm to be attended to. It is the timing of each step. This can only be established after a certain amount of practice.

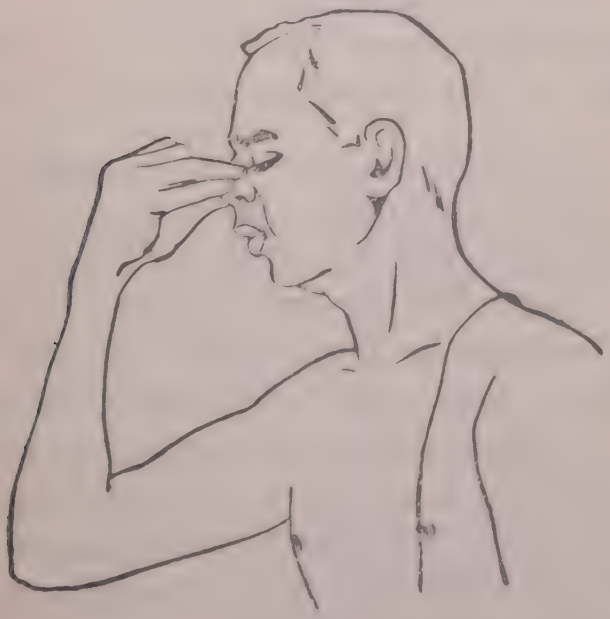


Figure 18

Exhaling residual air through the mouth

Initially one needs to concentrate upon Rechaka. The beginner will not be able to exhale his maximum capacity. Rechaka through the mouth will help to remove more of the residual air. The beginner may do it through the mouth as the first stage, first through one nostril and then through the mouth as the second stage, and completely through any one of the nostrils, alternately, as the third and final stage, when he is able to exhale to his maximum capacity through the nostril and when nothing more is possible through the mouth. This means that proficiency in Rechaka through nostril spontaneously eliminates its necessity to be completed through the mouth.

The need to practise Pranayama is greatly stressed by Hata Yogis, because their stress is on Aantarik Kumbhaka. But, in the Pranayama that we teach, the importance goes to Rechaka and consequently the danger of violence is completely eliminated. Therefore the warning regarding the need for an expert to supervise the practice by a beginner is wholly applicable to the traditional method which is prescribed in the books and not *this non-violent* method advocated in this booklet.

Even in improving the Rechaka, the pupil should be nonviolent. He must not aim at

sudden reduction of the residual air, especially if he is sickly and bedridden. By attention and by will, and by these alone, he can improve it.



Figure 19 The wrong way—the belly accommodating for inspiration, the chest remaining rigid.

A defect in our normal living can be markedly observed only in people with pronounced defects. Observe a person with a protruberant belly while lying flat on his back. It is seen that while inhaling and exhaling the belly goes up and down respectively. At the same time no movement is seen in the chest; the chest cavity neither expands nor contracts. These are proof enough of the fact that the thorax remains rigid and the expansion and shrinking of the lungs is achieved only by the diaphragm pressing down on the belly, resulting in what was observed earlier. This sets in an unidirectional dilation of the lungs. The amount

of the supplemental air and thereby also of the complimental air can be increased to a great extent, if the thorax retains or regains its elasticity.

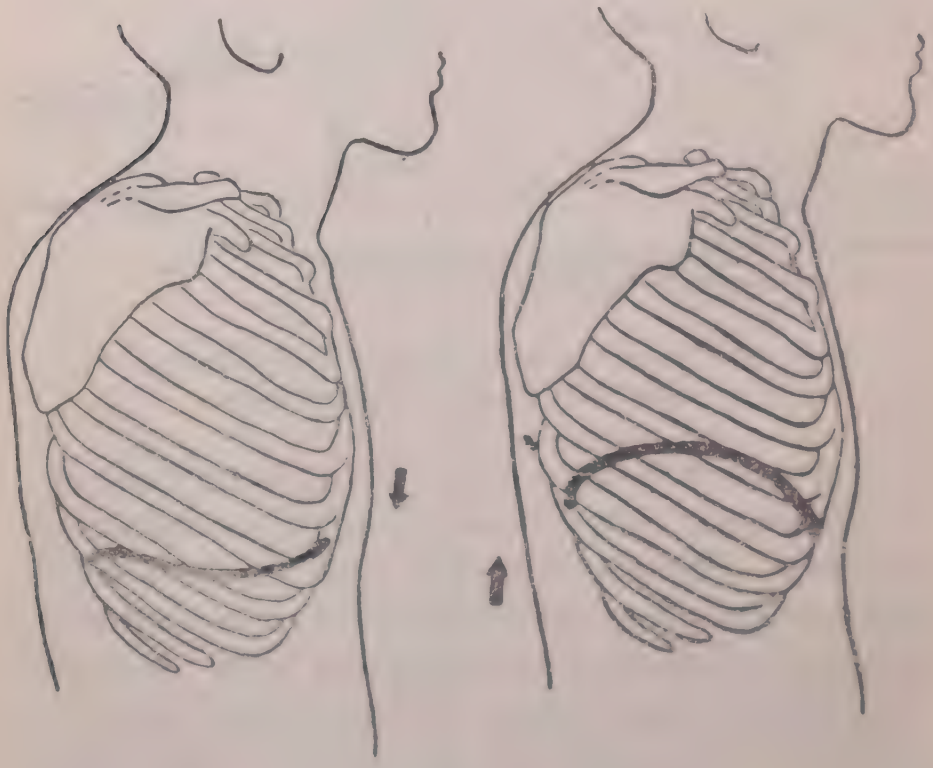
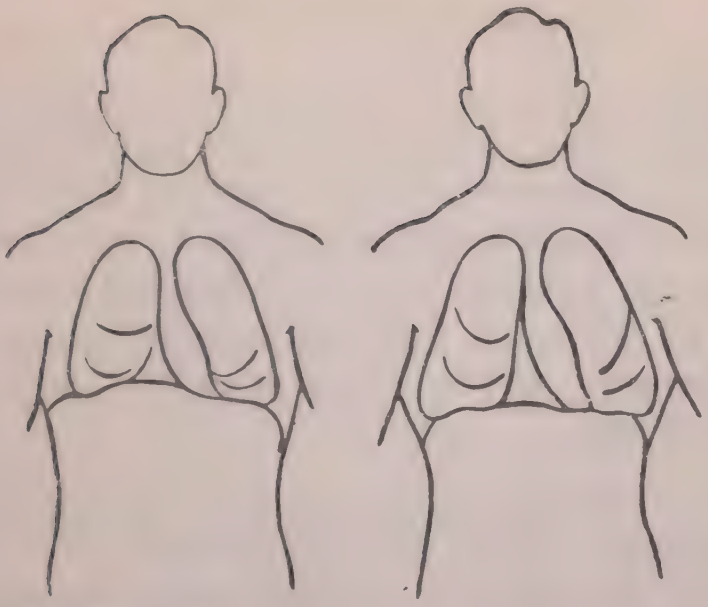


Figure 20

Effect of inspiration and expiration on the chest bones and the diaphragm—lateral view. Arrow indicates the direction in which the diaphragm moves.

Except in the case of people who have been doing exercises to develop the chest or in the case of people who have been practising



Figures 21 & 22. Effect of expiration and inspiration respectively on the lungs. chest and the diaphragm (frontal view). The lungs and the chest shrink and the diaphragm pushes up, during expiration. The lungs and the chest expand and the diaphragm is pushed down.

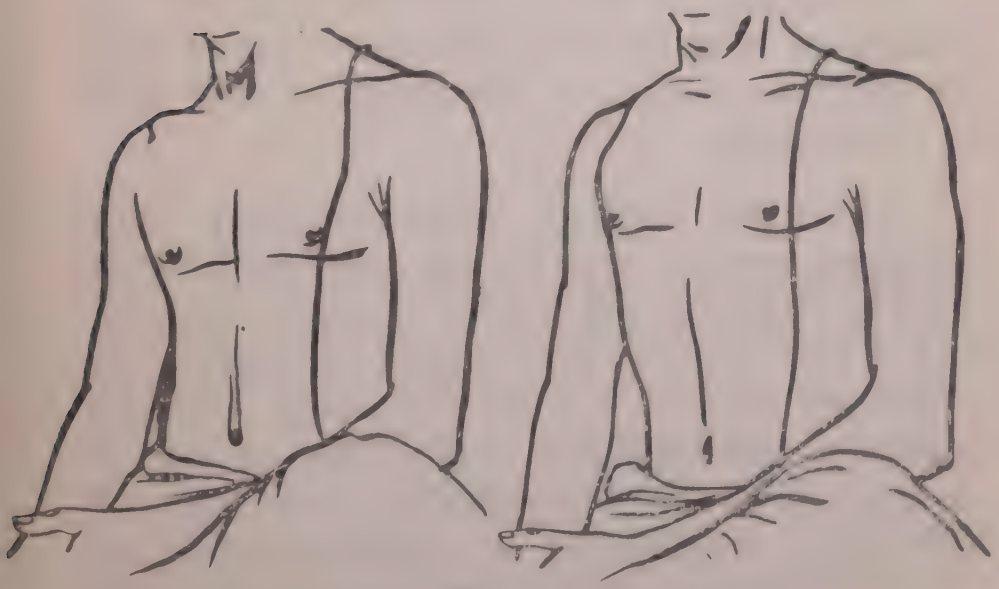


Figure 23. The chest in normal expiration
Figurs 24. The chest in normal inspiration

Pranayama in the right manner this elasticity is absent. An attempt should regularly be made to regain the elasticity of the chest by consciously breathing through the chest almost all the time. Such an attempt need not necessarily be confined to Pranayama alone, because the expansion of the chest improves normal breathing too.

There need be no special effort to improve the Pooraka. With the improvement of the Rechaka, Pooraka will automatically improve, for there will always be an attempt to equalise the air pressure and it will never be allowed to remain less.

Returning to the question of rhythm, the traditional method prescribes one unit of time for Rechaka, two units for Pooraka and four for Kumbhaka.

But in our method, where Rechaka is to be given greater importance, this one unit of time for Rechaka will not be enough. Another fact is that because the maximum possible air has been eliminated from the lungs through attentive Rechaka and that there is also a Baahya Kumbhaka following it, it may be difficult to double the time for pooraka. In the traditional method only one Kumbhaka

has been mentioned. But we have two kumbhakas, the Baahya Kumbhaka and the Aantarik Kumbhaka. To the beginner, it may be very difficult to allow four units of time for the Baahya Kumbhaka. It can only be attained after a perseverent, non-violent and long practice, along with dietetic righteousness as the main means of improvement of health. Until the practice of Baahya kumbhaka is improved the Aantarik kumbhaka need not be extended beyond the extent of Baahya Kumbhaka. The two can go at the same pace, just as Rechaka and Pooraka improve at the same pace. It will be enough to have one unit of time for Rechaka and Pooraka and one unit for both the Kumbhakas individually. If by practice one is able to increase the time unit for the Kumbhakas to two, three and even four, it will be advantageous because the Baahya Kumbhaka provides relaxation for such a long period to the lungs. We count the benefit only on that and not on the Aantarik Kumbhaka. The Aantarik Kumbhkaka is more to maintain the rhythm.

The following rules may also be followed in the practice. Before starting the practice the breather should make sure that his stomach is not heavily loaded, if it is not altogether empty;

the latter is a more preferrable condition. Of course no sincere pupil is likely to load his stomach heavily, as a rule.

Baths and exercises may also be employed for quickening the progress in the practice of right breathing. A Spinal Bath completed by a cooling bath for the head or by a piece-meal-cold-friction bath, according to the instructions given in Practical Nature Cure, or by a thorough wet friction all over the body, is a good preparation for the practice of rhythmic breathing. Any kind of mild exercise followed by any of the baths suggested here will also serve as a good preparation for a practice of Pranayama. Pranayama may also be practised during Spinal Bath or during Friction (Shallow) Hip Bath or while taking a particular exercise.

If Pranayama be practised during exercise, special care should be taken that it is non-violent. For example, in such exercises as are useful for chest expansion, there should be a breathing out when the chest is contracted and a breathing in when the chest is expanded. Also, in any exercise in which the lungs are subjected to pressure, the breathing in should be timed to the application of the pressure and the breathing out should be timed to the release

of the same. One thing more that we are most interested is that there should be pauses after each of the expansion and contraction, wherein the two kumbhakas can be fitted in.

The place chosen for practising Pranayama should be an open place. If such a facility does not exist, it must be one where there is a natural movement of cool fresh air, free from smokey fumes from mills or factories or other air-fouling agencies. A place near the window in a room with cross ventilation is suitable for the practice.

The breather should sit erect with his body naked above the waist. The dress worn at the waist, dhoti, pyjama or pantaloons should be loosened so that there is absolutely no pressure on the abdomen. It is good to have a large sized mirror in front so that the breather (the beginner) can watch if he sits straight and if the chest expands and contracts while breathing in and out respectively.

While practising, the pupil may meditate on the beneficence of Nature, and remember that he is getting better and stronger. This is not to be an auto-suggestion by compulsion, but an act on conviction. We

do not desire any one to accept our prescription just because of our authority, but because of the sound reasons as the basis.

The breather may use one nostril first, and then the other, in the following order. By the left nostril he should do Rechaka, the Baahya Kumbhaka following it when both the nostrils

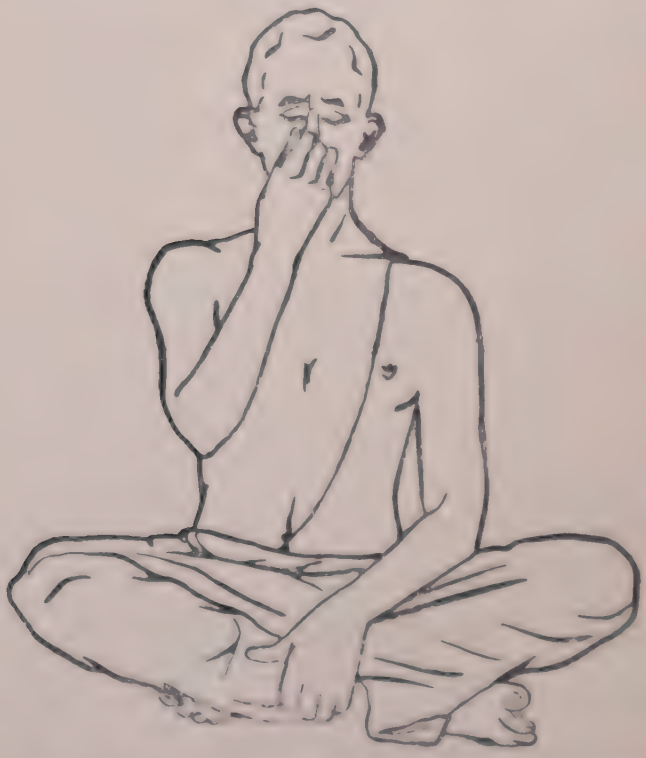


Figure 25. Sitting posture for the practice of Pranayama

and the mouth are closed, then by the other nostril Pooraka and Aantarik Kumbhaka following it; then he should do Rechaka by the right

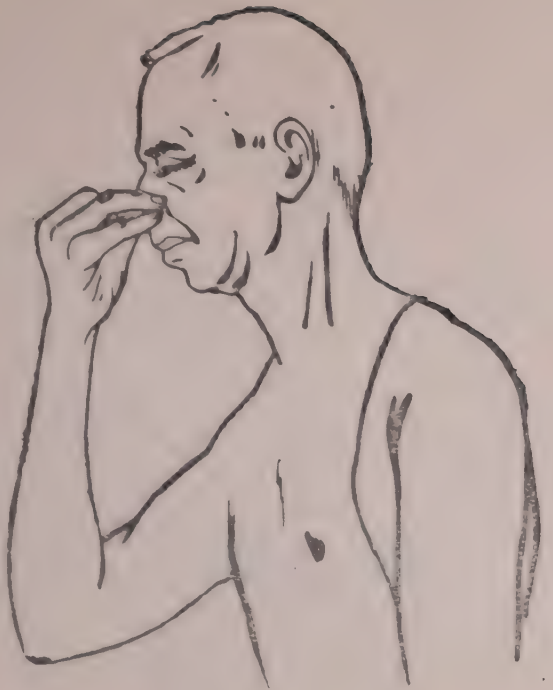


Figure 27 Baahya Kumbhaka

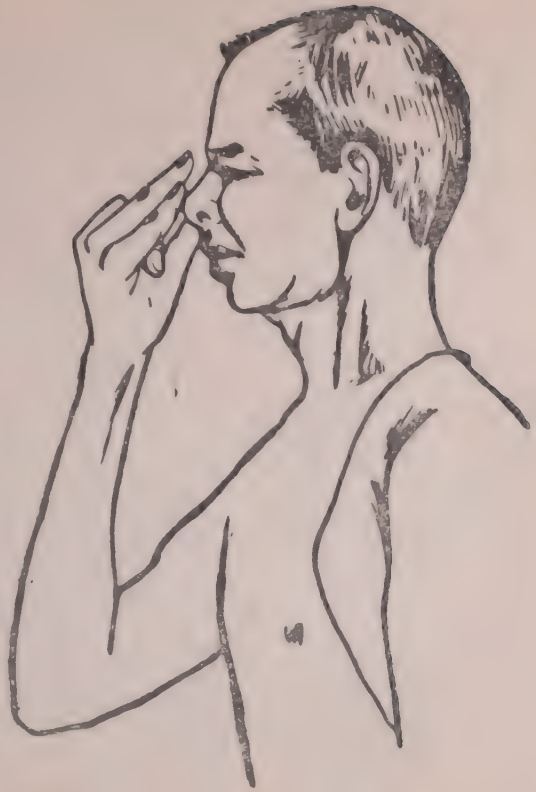


Figure 26 Pooraka through left nostril



Figure 28
Pooraka through right nostril

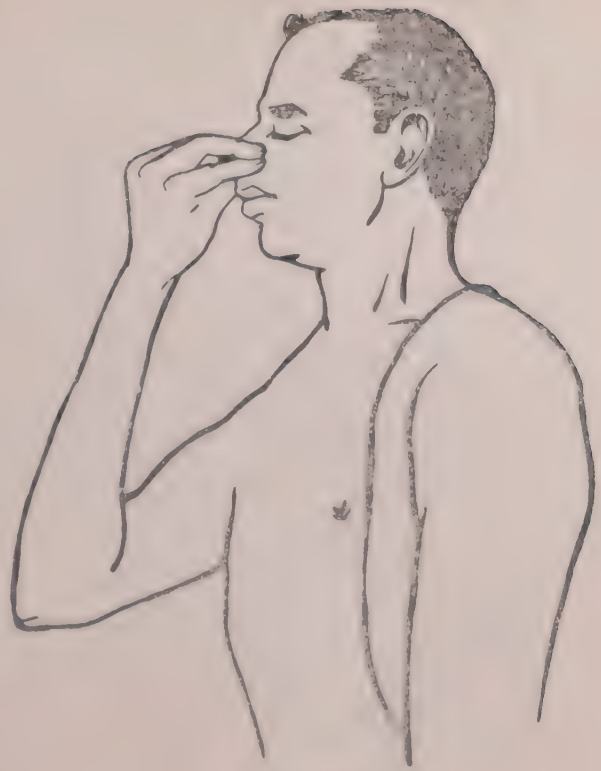


Figure 29
Aantarik Kumbhaka

nostril, again the Baahya Kumbhaka following it, and do Pooraka by the left nostril, Aantarik Kumbhaka, and so on. When one is in use, the other nostril should be pressed shut with the thumb or the little and the ring fingers, as shown in the picture (Figure 30).



Figure 30.
The mudra—the mode of holding the right hand fingers for the practice of Pranayama

The practice of Pranayama in this method should not cause any tiresomeness to one who has mastered it. But the beginner may feel so. He may rest after every few Pranayamas Gradually, he can increase the number of Pranayamas done at one stretch.

He may devote a few minutes to practise each time, repeating it two or three times a day, and afterwards increase the number of Pranayamas each time and the duration of each sitting. It will be good to have a certain

regular hours for the practice. He can even practise at odd hours, each time a few times, taking care that there is no heaviness in the belly.

Pranayama will confer appreciable improvement in health, both bodily and mental, from the beginning.

The bowel, stomach and other organs, especially the liver in the belly, are given a kind of massage and exercise by the regular lifting up and pushing down, which is all to the good. Both indigestion and constipation will be helped to a cure by this means. The foreign matter is burnt up and made into ash, which can be washed out with the help of the No-breakfast-Plan.

Pure cold air, if breathed in after a good Rechaka, will also help to heal up the sore lungs, the bronchi and the trachea, if any, as in cases of 'itis' (inflammation) of the pulmonary (pertaining to the lungs) system, like bronchitis, asthma, etc.

Pranayama is the best medicine there is for vocal musicians to improve their voice. It is with this point in view that they are advised

to keep up their regular practice of music in the early part of the morning, much before sunrise.

One may argue that practice of music is by itself a type of Pranayama. There is only one major flaw. The Pooraka is not done through the nostrils. If that is managed, then a musician does not need Pranayama so much. Also, there is no Baahya Kumbhaka possible to rest the lungs. Hence, Pranayama is essential for the musician, even as a corrective.

It has been observed in a few cases of partial deafness (for many have not so far come to our knowledge) Pranayama has helped greatly to improve hearing.

A good breather alone knows how to walk aright. Hence he profits by walking as an exercise more than others.

The effects on mental health are important. If the breathing becomes rhythmical, the mind becomes cool and steady. Even before that, improvements can be seen in many respects. Better breathing changes a negative unresisting mind into a positive, vigorous, and therefore self-assertive and happy one.

A good breather will tend to prevail on his fellows and competitors. He becomes a better leader of men than he can otherwise be.

6. NOTES ON SLEEP

Sleep, in the sense of consciousness, is not a vital need but only relaxation of mind and body. Animals, especially wild ones, do not sleep as we do, but only relax perfectly. Hence sleeplessness is not such a serious evil as it is made out to be.

Relaxation comes naturally with honest labour, dietetic righteousness, and a healthy temper of mind. Sleeplessness in its worst form is a disease of the mind. It can be cured only by diet, fasting and intelligent self-surrender. [7]

It is not in our power to go into the state of sleep by our own will or effort. Neither sleep nor death are acts of the will. They happen by the intervention of an over-riding power which the west calls Nature, but which is just the all-pervading Power of God. Sleep comes by the Grace of God. All that we can do is to live in such a way as to deserve grace. There is no other way.

7 In this connection the reader will do well to go through the chapter "Mental Healing" in "Practical Nature-Cure".

The idea that sleeplessness is a difficult problem to be solved by specific expedients,—without a general reform of the habits of life—is an absurdity, that can be pardoned in the doctors, but not in the followers of Nature's way.

The more one tries to sleep, the more awake one becomes. It is the wiser part to live as to deserve sleep and then to leave it to Nature instead of worrying.

The mere loss of sleep is not so harmful as the worry about not getting it. The evil effects that are usually debited to sleeplessness are really due to the worry that the sleepless one is guilty of. In the matter of sleep, as in all things else, we must take the hint given us in the Gita, "कर्मण्येवाधिकारस्ते मा फलेषु कदाचन।" "Thou hast the right only to do the right actions, not to desire the fruits thereof". We need to resign ourselves in advance to the Will of God as to the consequences of our actions. By such resignation one gets far better results than by thinking about the results and worrying. Renunciation of the fruit of actions is the best part of Renunciation, and thus renunciation has an immediate reward, peace. The Gita says: "त्यागाच्छान्तिरनन्तरम्" "After renunciation peace comes at once".

The less the food, the less the sleep needed. Sleeplessness injures the abstemious very little, or not at all. Let the sleepless note this. Yogis are able to keep well with little or no sleep because of their frugal eating and renunciation of the results.

When it is known in advance that one will have to forego one's sleep of night, one would do well to miss that meal, so that one might not need that relaxation, in view of the fact that food is a tax on vitality.

But when one had to miss the sleep after one had one's meal, one might naturally try to compensate for it immediately one finds time or leisure. Care must be taken to have that compensatory sleep on an empty stomach, so that sleep does not go to pay the tax of the food but to compensate for the loss of sleep only. Otherwise, however much one might sleep, missing none of the regular meal programe there will be no feeling of having had any relaxation; one will not feel refreshed on walking; the drowsiness will continue.

One may be ready to miss a meal, but if there is not time enough to compensate for

the loss of sleep, a sleep in the Spinal Bath will equal at least four times the sleep in bed.[8]

In illnesses the surest way to get sleep is to take to the Nature's Way, since this cure is essentially a surrender of the case to Nature. And to the sick one sleep is the chief medicine there is. Fasting is another.

A rhythmically ordered life,—in tune with Nature—makes sleep unnecessary.

Let the sleepless man meditate on the truths of Religion, the nature of God and his own Inner Self, and the relation between them, according to the teaching of his own religion. Let him thus forget his body and all worldly concerns.

Or, let him meditate on the Name and Form of his Ishtam (liking)

The mere effort to meditate will bring on sleep, except in the case of adepts in the practice of meditation. Success in meditation being what it is, there is no better reward and there is no harm if sleep does not come; it is still better.

8 Read the book "Spinal Bath For Nervous Tone" by L. Ganesa Sarma.

In a case of sleeplessness, the sufferer was advised to live upon an exclusive diet of vegetables, eschewing rice, and that taken once a day. By observing this plan the sufferer got rid of his trouble within a month and he is continuing with a single principal meal and another of only vegetables or fruit. Addition of rice to his diet upsets his health, and he has given it up.

In the year 1949 the author's son, Ramachandra Sarma, treated a complicated case when he was at Alwar. The gentleman was himself a druggist and the father of a medico. After eight years of attempt at all drug treatments by well-known experts and having had not a wink of natural sleep, he chose the natural way. Immediately on admission in the sanatorium he was given an enema to clear his heavily constipated bowels to some extent. Then he had a hot immersion bath followed by Spinal Bath. The very first day he had a taste of natural sleep, though for only a short while, after eight years of yearning. Gradually, with dietetic reform all his troubles were relieved and sleeplessness too.

The following further hints are given for the benefit of the sleepless.

Go to bed with a stomach almost quite empty. If supper be taken at all, it should be early and very light. (If you are on a single meal plan and if you take it after the day's work is over, you need not so much bother about the stomach's becoming empty.) At first this may even make it worse. But do not mind it. Such sleep as one gets on this plan will be more restful so that less of it would suffice.

Walk up and down, slowly and mechanically, thinking of nothing, till nearly tired and then lie down and wait calmly for sleep.

Or, take a warm immersion bath for the whole body, from the neck down, for about fortyfive minutes, finishing when the water ceases to feel warm. Then finish with a cold wash or the head. Dry up and lie in bed, well covered.

Or, take a cold foot-bath for about ten minutes, and then either warm the feet under a blanket in bed or by a hot foot bath. The first foot bath must be very cold, and the feet must be dipped up to the ankles. During the bath the rest of the body may be well wrapped up.

Or, inhale a deep breath with the least possible effort, and then allow the inhaled air to go out very very slowly. Repeat this till you become drowsy.

When waiting for sleep, imagine that you are no longer in the body; let it lie in the bed like a lifeless thing. Let the breathing go on for itself without effort. Unclench the teeth and loosen the joints.

Sleeping too long is a source of disease. It is better to rise as soon as refreshed. Enough is far better than excess. Especially is this the case with the midday nap, where this is needful; more than ninety minutes' sleep will lower the vital powers.

APPENDIX—I

CURE OF ASTHMA THROUGH THE LIFE NATURAL

Asthma [9] is a weakness of the breathing mechanism, occasionally resulting in a near approach to suffocation. It is not by itself a fatal disease, unless interfered with drugs. Efforts to relieve the symptoms of asthma with adrenaline, ephedrine, etc. for relaxing the muscular walls which have caused a constriction of the bronchial tubes, or with histamines[10] to relieve the swollen condition of the mucous membranes of the bronchial tubes, are in the long run a very bad policy; for the disease, thereby, becomes firmly rooted in the constitution. In most cases the fault is not so much in the lungs, but primarily in the digestive organs, in the nervous system and, to some extent, in the mind.

---

9. Asthma (Greek-a panting) Paroxysmal attack of difficulty in breathing. Bronchial a. attack of breathlessness associated with bronchial obstruction or spasm. Characterised by expiratory wheeze,. Cardiac a. Paroxysmal dyspnoea in left ventricular failure. Status Asthmaticus, repeated attacks of asthma without any period of freedom between spasms.

10 Histamine--imidazolethyl amine--produces vasodilation, lowers the coagulability of the blood, etc.

The basic cause is the same as for any other chronic disease, namely sub-normal health due to wrong living, chiefly an uncontrolled appetite for savoury edibles, highly spiced and otherwise appetising foods such as fried stuff. Because of this cause, the asthmatic will never become free from the disease, unless he learns self-control. It is not enough to get a cure. It is also necessary to stay cured by cultivating positive health for the remainder of one's life. And let it be noted that health must be deserved in order to be won and retained. The only real cure of any disease, especially chronic ones, is the regaining of lost health. There is no other way.

The asthmatic must begin with a fast of one to three days, thus giving to life relief from the usual burden of digesting and assimilating food, and of excreting the residual waste. This enables the life power to deal with the urgent work of repairing the digestive organs and making them fit for their work. This is all the more necessary, because in any chronic case the digestion is more or less defective, so that the food, instead of promoting health, actually promotes disease.

After this short fast the patient should take to a course of highly positive diet, limited in

quantity, so that the maximum benefit can be achieved with the minimum expenditure of vital power, for, what carries on the work of cure is not diet but the vital power. The less one eats, the greater will be the power available for continuing the cure, which is begun by the fast. The test of vitality is the feeling of lightness and general bodily vigour.

The most highly positive of all foods are fresh, green Saattvic herbs, preferrably the wild grown ones. Hence juice of such herbs, in minimum doses diluted with water, should be regularly taken at least once a day on an empty stomach after a sense of hunger is manifest. [11] This is not only food but also the medicine to neutralise the evils of the previous over-eating of negative foods. As an alternative, juice of ashgourd (ash-pumpkin, பூசணிக்காய் पेठा) extracted by squeezing after grating, can also be taken in a dilute form. In the case of adamant people or children, who have to be coaxed to take it, it can be added to a small quantity of tomato juice, carrot juice, pomegranate juice, very thin buttermilk or sweetened with a spoon of honey.

11 Refer to chapter on 'Vital Economy', in 'Practical Nature-Cure' by the same author.

Vegetable soup or juicy fruits (of only one kind at a time) may be taken at other times during the day. If juice of fruits is desired, instead of the whole fruit, it *must be diluted*, as otherwise it may cause overfeeding, and rapid consumption. Of course the sharp taste will be made milder and that will, in turn, compel it to be taken more slowly—eaten and not drunk. Each time there must be natural hunger before the diet is taken. The quantity taken must also be limited; so as not to cause the least sensation of heaviness.

Basking in the sun, light exercises of some kind, Yogic Asanas and Pranayama—even ignoring Aantarik Kumbhaka—are also both necessary and highly beneficial. Hydro-therapic baths [12] are also useful. The mind must be purified by any suitable means suggested in philosophical and religious scriptures. That helps to remove the strain due to worry which retards recovery. It is good to think of the righteous path that one should take. But *it is wasteful to worry about the unpalatable past*. Even about the future, when one is not well, it is better to shift the responsibility on a reliable person, one who is known to pursue the

12 Refer to "Spinal Bath For Nervous Tone" by L. Ganesa Sarma.

Path of Truth. The patient must be always optimistic about his cure; for the Nature's Path assures that.

Since vitality is depleted by all kinds of self indulgence, among them the sensual ones, self-restraint in all these respects is an essential part of the cure. This self restraint in eating and other ways of wasting vitality, what we call Vital Economy, is a good deal more than half the cure and that neglect of it is very likely to make the cure impossible. He that understands this principle as well as the law of positive-curative dieting, is qualified to be his own doctor for the rest of his life.

The habits set up in the cradle carries on for life, for good or otherwiee. The mistaken notion that children should eat as much as they can manage to digest creates a foundation for chronic diseases in later ages. The constipation, that is uncared for, helps to deposit the morbid matter in different parts of the system and in some children the accumulation occurs in the lymphoid tissues of the adenoids, by the thickening and blocking of the posterior nares (the pair of openings from tha nasal cavities into the naso-pharynx); as a result, they find difficulty in breathing through the nostrils and

do so through the mouth. Gradually, this deforms the face with the lower lip hanging down and thick. As age advances, without any reform in the wrong ways of living, they become victims of chronic diseases, like asthma.

Common cold is often a trouble for which the ' change in the weather ' is accused. There are people who have it as their permanent companion. Medical people agree that it is incurable. One famous doctor observed that 'left to itself it will be gone in seven days, and treated medically it will take a week to cure it.' But still people do take some drugs or other for relief from the discomfort, not realising the wastefulness of the steps taken. Some people in India plug their nostrils with snuff (tobacco powder) for relief from cold, but actually confirm cold.

As the allopathic and other drug treatments are symptomatic, the side effects of catarrh (nasal), some of which are listed below, are treated as ones having no connection with the real cause :-

1. Earache, or inflammation of the mastoid bone, by the extension of catarrh to middle ear through the eustachian tube;

2. Frontal headache, due to extension of the catarrh in the frontal sinuses;

3. Pain over the top of the head by the extension of the catarrh to the sphenoidal sinus;

4. Difficulty in swallowing (dysphagia) through inflammation of the mucous membrane from the posterior nares on to the pharynx;

5 Forcible expulsion of tears by reflex action; and

6. Inflammation of the lymphatic vessels and glands in the oral cavity.

When the very cause of these troubles is confessedly incurable, how can these side effects be cured? The relief obtained is only a self-deceit and one that paves the way to lasting troubles.

Detailed instructions for the treatment of all ailments are to be had from " Practical Nature–Cure " (referred earlier) under the relevant chapters as demanded by the trouble according as it is either acute or chronic.

Maltreated or ill-treated, acute troubles have their obvious progression to destructive or degeneracy stage like tuberculosis (consumption) or cancer through the chronic stage like asthma. Therefore, it is not a wise policy to bank disease for the future.

Nature's Path is the God-given wisdom by which health and cure are deserved and won.

-o0o-

APPENDIX II

FROM THE VEDIC LORE

Readers of our text-book, Practical Nature-Cure, do know that the Life Natural is a Science with a strong foundation, evidence for which is obtainable in our ancient lore, the Vedas and the Upanishads. Doubtlessly, Pranayama is equally so. The Life Natural has for several centuries been forgotten and had to be re-discovered. But the tradition of this practice of Pranayama has been kept up by the Brahmanas as an important part of the daily Sandhya-vandana. It is a regrettable feature that in its practice, the essential part of it has been ignored by most of the people and just the repetition of the mantra has been taught. Perhaps, it was due to the fear that the technique of Pranayama needed to be taught and the practice supervised only by an expert and the purohits were far from being experts, and therefore preferred to be safe, relegating that aspect to the class of experts known as "Yogis", whom only those who renounced the world sought after.

The element of danger that was shadowing this sacred practice of Pranayama has been eliminated in this book. Hence, such of those as have had the initiation in this mantra according to the ritual method, can henceforth commence the practice with the repetition of of the mantra in a proper order.

In addition to the instructions given here-to-before, a little that has been culled from the ancient texts is given below, eliminating such details as refer to the popularly known, but violent type of Pranayama.

It is said that in a day an adult breathes 21,000 times. Every breath saved adds to the span of life. So,the practice of Kumbhakas(both Baahya and Aantarik) is essential for the longevity of life. This is a simple explanation for the necessity for the practice, in addition to the other and major physiological ones explained in the foregoing chapters.

Of the five Primary Elements, Ether, Air, Light, Water and Earth, though all of them provide nutrition for the living body, the finer ones serve as cleansers as well. The cleansing quality ascends with the fineness of the Element. Therefore, Air is a better cleanser than the following ones, viz., Light and Water.

For the sustenance of any Element, the presence of the foregoing ones is essential. So, when Ether and Air are there, Light or Fire also can exist. And we know that these three are cleansers.

Just as we purify gold by the use of Fire, the dirt in the system is also consumed by the Fire which is sustained by the Praanavaayu (oxygen) that we breathe in larger quantities by better breathing.

Accumulation of filth in the system is due to physiological sins. And, as we have a means of burning it out by Pranayama, we are able to expiate our sins. It is perhaps with this reason that the "Kalpasutram" prescribes Pranayama as a "Praayaschitta" for the sins. Bodily purity helps in the attainment of mental purity in turn. One will therefore be able to atone for moral sins as well.

The same is said in the Commentary by Bhagavadpaada Sri Sankaraachaarya for the "Svetaasvataropanishad";

प्राणायामक्षपितमनोमलस्य चित्तं ब्रह्मणि स्थितं भवति.....

"He, whose sins are depleted by the practice of Pranayama, has his mind fixed in Brahma....."

"Pranayama comes only after 'Yama' and 'Niyama'. Yama, according to Patanjali's Yoga Sutraa (Chapter II, sutra 30), consists in not killing (अहिंसा), truthfulness (सत्य), not stealing (अस्तेय), continence (ब्रह्मचर्य), and not accepting gifts (असंग्रह); and niyama (Chapter II, sutra 32) in internal and external purification, contentment, penance, study of the Vedas and worship of God." (Sandhyavandana, by Sri B. V. Kamesvara Aiyar, Pudukkottai).

The "Svetasvataropanishad" and its commentary by Sri Sankara, describe the posture, etc., for the practice of Pranayama:

त्रिरुन्नतं स्थाप्य समं शरीरं इन्द्रियाणि मनसा संनिवेश्य ।
ब्रम्होडुपेन प्रतरेत विद्वान् स्रोतांसि सर्वाणि भयावहानि ॥
Chapter II, Mantra 8

"The head and neck and chest being held straight in one line (see Figure 25) through conscious effort fixing the senses in the heart (not the material heart), sailing in the boat of Brahma (ब्रह्म शब्दं प्रणवं वर्णयन्ति — Sri Sankara's commentary says that the word 'Brahmam' stands to denote the 'Pranava', 'ओं'.), the wise one crosses the ocean of all miseries." (In the commentary, Sri Sankara elaborates the miseries of life that are born through Ignorance,

Desire and the (past) Actions, causing recurring death and birth in human, animal and plant forms; freedom from all of which is attained by the wise.)

आसनानि समभ्यस्य वांचितानि यथाविधि ।
प्राणायामं ततो गार्गी जितासनगतोभ्यसेत् ॥

"After well practising the aasanaas of your liking in the proper manner, (Gaargi,) then do Pranayama in that aasana (pose) that you have mastered."

Though earlier it has been stated that the pose should be one in which the chest, neck and the head are held erect in a line, Pranayama is also prescribed in any of the sukha-aasana that has been mastered by the follower, as a part of the yogic or other exercise. (See chapter on Practice of Breathing). It is for this purpose that the student is advised to practice well the asanas of his choice and that too in the proper manner, find out the one he has mastered and utilise that for the combined practice of Pranayama along with the asana.

मृद्वासने कुशान्सम्यगास्तीर्याजिनमेव च ।
लम्बोदरं च सम्पूज्य..... ॥

"Being on a soft seat with darbaasana and deer skin and after worshiping Sri Ganesa....."
(Worship helps to cast oft worries.)

तदासने सुखासीन..... ।
समग्रीवशिरः संवक्त्रसंवृतास्याः सुनिश्चलः ॥

Sitting comfortably with the head and neck held erect,[1] and covering the body (with a cloth to avoid any possible disturbance by small insects like the flies or the mosquitoes) so that the body remains motionless,

प्राङ्मुखो वापि नासाग्रन्यस्तलोचनः ।
अतिभुक्तमभुक्तं च वर्जयित्वा प्रयत्नतः ॥

Thus seated, facing east or north, the eyes looking at the tip of the nose, and seeing to it that (you) are neither over-fed or unfed (you) must practise (Pranayama)

Taking this lesson, many Hatha Yogis advise that Pranayama should not be practised on an empty stomach. They advise therefore a breakfast of milk and fruits before practice. This is definitely contradictory to the general diction. No food can be had before worship is

1 The same instruction is given in the Bhagavad Gita (Chapter 6, sloka 13)

समं कायशिरोग्रीवं धारयन्नचलं स्थिरः ।
संप्रेक्ष्य नासिकाग्रं स्वं दिशश्चानवलोकयन् ॥

over. We saw here that worship of Sri Ganesa is prescribed as a preliminary to Pranayama. Then how can there be any food before that? It is therefore a clear mistake and naturally such studends as follow the advice, that there should be a breakfast of milk and fruits before the practice of Pranayama, either reap no benefit by the practice or lead themselves to woe.

There is another mantra that guides the basic principle of Health and Healing, which says (अन्नं हि भूतानां ज्येष्ठं तस्मात् सर्वौषधमुच्यते।) 'Food is elder to beings and hence the Supreme Medicine. it is said ". This tells us that there is a principle of respect for the elder one and that the younger one can sustain only in the presence of the elder. Applying this principle, we see that it is the second of the Five Primary Elements that is concerned in Pranayama. Obviously Ether, the elder Element to Air, should also be present. Ether is available only when the stomach is empty. It naturally follows that Pranayama should be practised before the meal and not after it.

But we do not hasten to condemn this teaching of Sri Sankara in his commentary to the " Svetaasvataropanishad ", for the real meaning is that, since complete rest is advised

on days of fasting. there should be no practice of Pranayama on days of long fasts. Pranayama is, to some extent, an exercise, and it is proper that it should be avoided when there is to be fasting for that day.

This rule equally applies to fasting due to illness, when also Pranayama should be suspended.

Otherwise, Pranayama should be practised on an empty stomach. During the three occasions of worship, viz., प्रातः (morning) सन्ध्या, माध्याह्निक (midday) सन्ध्या, and सायं (evening) सन्ध्या (Sandhya), the aspirant is expected to have his stomach empty. The rule is that there should be no breakfast, as is generally understood. The afternoon meal, the principal meal is had only after the midday Sandhya, which is followed by rest. Obviously, by the time the evening Sandhya is due to be performed, the stomach would have become empty. And the rule prescribed by Sri Sankara will strictly be adhered to.

दक्षिणनाशिकापुटमंगुल्यावष्टभ्य वामेन वायुं पूरयेद्यथाशक्ति। ततोऽनन्तरमुत्सृज्यैवं दक्षिणेन पुटेन समुत्सृजेत्। सव्यमपि धारयेत्। पुनर्दक्षिनेन पूरयित्वा सव्येन समुत्सृजेत् यथाशक्ति ॥

"Closing the right nostril by pressing it with the thumb, *inhale* through the left nostril

as far as you can (not to the extent of straining). Afterwards, release the air through the right nostril. Inhale through the right (nostril) also. Again inhale through the left nostril and exhale through the right, *as far as you can.*"

The word यथाशक्ति (as far as you can) occurs in both contexts, viz., Pooraka and Rechaka. We have two lessons from Sri Sankara here; i. The exercise should not be carried to the extent of straining. This keeps off violence and danger ii. Rechaka should be carried to the limit possible. In Rechaka the possibility of danger is only when the lungs are completely emptied to cause a real vaccuum inside. But that is absolutely impracticable, and danger cannot be there at all through Rechaka.

Pranayama is classified as सगर्भ प्राणायामः (Sagarbha Pranayama) and अगर्भ प्राणायामः (Agarbha-Pranayama). When performed with chanting of the mantra, it is known as Sagarbha-Pranayama, and when it is done without the mantram as Agarbha-Pranayama. As a ritual, it may be practised as Sagarbha-Pranayama and as an exercise for the attainment of cleanliness of the physical body and thereby also of the mind, as Agarbha-Pranayama.

The Sagarbha-Pranayama is expected to be performed, as minimum, 45 Pranayamas a day split in three occasions, morning, noon and evening. They are: 1 in Sankalpa, 1 in Argya-Sankalpa, 1 in Aaditya-Upaasana, 1 in Japa Sankalpa, 10 before Gayatri Japa and 1 before Gaayatri-Upasthaana.

ओं भूः ओं भुवः ओं सुवः ओं महः ओं जनः ओं तपः ओं सत्यं ।
ओं तत्सवितुर्वरेण्यं भर्गो देवस्य धीमहि । धियो योनः प्रचोदयात् ।
ओमापोज्योतीरसो ऽमृतं ब्रह्म भूर्भुवस्सुवरोम् ॥

— —

IMPORTANT NOTE

Initiation in any mantra has to be through a Guru (preceptor) and in its chanting the pronunciation, accent and metre should be properly observed. It is given here ONLY for the benefit of the initiated ones, who can combine understanding with dry practice.

This is a combination of four mantras, viz., (1) the प्रणव (Pranava), (2) the व्याहृतीनाम् (Vyaahritis), the गायत्री (Gaayatri) and the गायत्री शिरम् (Gaayatri-Siras).

The Pranava denotes the Suddha Brahman (शुद्धब्रह्मम). Since the seven (भूरादि सप्तव्याहृतीनाम) Vyaahritis are named preceded each by the Pranava, they denote the seven worlds created by the Almighty, and which are not beside Him. According to the Vedantic tradition, there are seven worlds above and seven below. In this mantra one set of them alone are named. But when the upper ones are named, the lower (पातालादि) ones are also implied thereby. By the repetition of the Gaayatri, we get strength of intellect. And the Gaayatri Siras is for the meditation of the Brahman in the forms of Water, Light and other Elements.

THE PRANAVA

The Vedanta speaks volumes about the Pranava. Here let us rest content with just the following explanation on it.

निर्गुणं निराकारं निर्विशेषं अखण्डं निरुपाधिकम् ।
एकमेवाद्वितीयं स्वतन्त्रं नित्यमुक्तं परिपूर्णम् ॥

"Brahman is attributeless (Nirguna), formless (Niraakaara), without special characteristics (Nirvisesha), unsplit (Akhanda), without any limiting adjunct (Nirupaadhika), the ONE without a second (Ekam-Adviteeyam), free (Independent) (स्वतन्त्रम्) ever-free (Nitya-Muktam), and ALL FULL (Paripoornam)."

THE VYAAHRITIS

"Bhuh, Bhuvah, Suvah, Mahah, Janah, Tapah, and Satyam" are called the seven Vyaahritis (Sacred utterances). They are the names of the seven (upper) worlds or spheres. "More often, as for instance at the close of this very mantra, and the invocations in the Yajur Veda, the first three alone constitute the

Vyaahritis, Suvah[2] denoting comprehensively all the higher planes of life from Svarga (Indra's heaven), to तद्विष्णोः परमं पदम् (tad vishnoh paramam padam), the highest abode of the blest, where the gods and the liberated rejoice for ever. Bhuh is this earth, where men and

2 There are two prevelant spellings for this one term. Generally, in the Southern parts of the country it is spelt as you see here (सुवः). In the north it is spelt as Svaha (स्वः). Readers should not hasten to condemn any one in favour of the other. In the Rig Veda and the Sama Veda it is spelt as is in practise in the North (स्वः) In the Yajur Veda also there are five specific references to this word under the portion known as Kaathakam known as Prasnas where the spelling is the same (स्वः). In other places in the Yajur Veda the spelling is Suvaha (सुवः). Followers of Krishna Yajur Veda adhere to the spelling of Suvaha.

In the Sri Vishnusahasranaamam there is a sloka which reads as follows:

भूर्भुवःस्वस्तरस्तारः सविता प्रपितामहः ।
यज्ञो यज्ञपतिर्यज्वा यज्ञाङ्गो यज्ञवाहनः ॥

In the first half of the first line the third word is स्वः If it be changed as सुवः the metre will become longer by one maatra and hence wrong. Here the usage is स्वः Svaha.

Thus we see that both usage is common to denote Svarga in this chain of the main Vyaahritis that mean: This world, the Antariksha and Svarga.

the lower beings are born or re-born to enjoy the fruits of their former deeds and where, by the way they aquit themselves, they shape for themselves their future destiny in the higher or the lower planes of existence." (Sandhyaa-vandanam, by Sri B. V. Kamesvara Aiyar)

GAAYATRI

The Gaayatri mantra was given to us by Sage Visvaamitra. It is understood to be the cream of Rig, Yajur and Sama Vedas. This also embodies the Pranava and the three Vyaahritis.

Gaayatri is the name of the metre of 24 syllables. Counting in the normal way, this mantra contains only 23 syllables. To make it 24 some people advise the elongation of one of the words वरेण्यं (varenyam) to वरेणीयं (vare-neeyam). That would mean something like cutting off of the foot to suit the shoe. In the Soorya-Ashtaakshari mantra the word "Surya' (सूर्यः) is counted to have three syllables. Similarly, here, in the Gaayatri mantra also, the word (Varenyam) is to be counted to consist of four syllables.

There are different versions of the meaning for the mantra. The popular approach is that

the prayer should be addressed to Devi (feminine), so that we could deserve Her (Gaayatri in Her three forms, Gaayatri, Saavitri and Sarasvati) compassion. Each one can assume any sense according to his own state of spiritual attainment. It is, in fact, a prayer to sun-God, सवितुः who is the manifest form of the Supreme Self.

The meaning in essence is this:

> "We meditate on the Supreme Glory of Him (Siva or Naaraayana—or Her Gaayatri, Saavitri, or Sarasvati) Who (in the form of the Sun) inspires our knowledge and is the Creator of the Universe."

GAAYATRI-SIRAS

It is so called because it stands at the head of the mantra. The Siromantra means:

> "The universe which consists of the Bhuh, Bhuvah and Suvarlokas and which in its primary Elements, consisting of Light and Water and the solid matter (Earth) is the Eternal Essence."

THE PRACTICE

Because we have four stages in a single Pranayama, viz. Rechaka, Baahya-Kumbhaka Pooraka and Aantarik Kumbhaka, we need to split the mantra into four parts for chanting at the time of the practice. As it is, the mantra consists of only three parts, since the Pranava and the Vyaahritis go together. We would prefer to leave it to the convenience of the student to use the mantra suitably. But, yet we would prefer that no individual part of it be split up. This would therefore necessitate the performance of any one step without mantra or just Pranava alone. It may be suggested that the Rechaka be done with just the Pranava alone, the Vyaahritis preceded by the Pranava follow with the Baahya Kumbhaka, the Gaayatri with the Pooraka and the Gaayatri Siras with the Aantarik Kumbhaka.

दिक्कालाद्यनवच्छिन्नान्तचिन्मात्रमूर्तये ।
स्वानुभूत्येकमानाय नमः शान्ताय तेजसे ॥

—भर्तृहरिः

"Prostrations to that Pure Light, whose form is Chit, Absolute, unconditioned by Time, Space, etc., whose sole test is direct perception."

Barthrihari

APPENDIX III

वस्त्रधारणम्

DRESS

प्राणहिंसा न कर्तव्या कथंचन कदाचन ।
व्याप्नोति नियमोऽयेष सर्वं शास्त्रमिदं शिवम् ॥ 1

'There should be no *violence* to Life in any manner and at any time' — this rule is implicit in every part of this great Science.

यथांगानां न हिंसास्यात् कर्तव्यं वस्त्रधारणम् ।
यथा च वायुसंबन्धो न बाध्यते त्वचा सह ॥ 2

Dress (therefore) should be so worn that there will be no violence (by pressure) to any internal organs, and also so as not to cut off the access of air to the skin.

विभजन्त्युदरं द्वेधा मनुजा वस्त्रधारणे ।
पीडैवमौदरांगानां स्वास्थ्यहानिरतो ध्रुवे ॥ 3

Men (and women) divide the belly into distinct halves by the way they put on their (under) clothes. Thus there is pressure on the organs in the belly and consequent loss of health, which are inevitable effects (of this habit).

मलाशयस्य दौर्बल्यं रोगाश्च विविधास्ततः ।
अनेनैवाचारेण भवन्तीत्यवगम्यताम् ॥ 4

Just by this mistaken habit there is weakening of the bowel and a variety of disease resulting therefrom. This has been clearly demonstrated (by workers like Dr. Kellog).

वसनं धारयेत् तस्मात् समग्रादुदरादधः ।
न कस्याप्युदरांगानां पीडनं भविता यथा ॥ 5

The wise follower must wear his dhoti below the whole of the belly, so that there will be no pressure at all on any of the organs in the belly.

स्त्रीणां विधिरयं मुख्यो माननीयो विशेषतः ।
अन्यथा ता भजन्त्येव दुःखं हि प्रसवे शिशोः ॥ 6

This rule is of special importance for females, and they should respect it accordingly. Otherwise they will suffer when giving birth to children.

रन्ध्रैर्वस्त्रगतैरेव वायुमङ्गो यतस्त्वचः ।
नीरन्ध्रं निविडं वस्त्रं परिधेयं न धीमता ॥ 7

Since the skin has access to the outer air only through the meshes of the cloth worn, the wise one should not wear close textured

cloth, in which the meshes are closed up. (Porous cloth, such as Khaddar, is recommended).

नाधिकं वस्त्रं धार्यं मितं पर्याप्तमेव तु ।
अधिकं धारयेत् वस्त्रं वायुरर्कभ्यां निरुध्यते ॥ 8

No excess of clothing should be worn, but only what is absolutely necessary. By wearing excess of dress the body is deprived of access to air and sunlight.

वर्णेषु कालिमा हेया रवितेजो निरोधनात् ।
उत्तमो धवलो ज्ञेयस्तयोर्मध्ये भवन्ति च ॥ 9

Because, of the colours, black (dark) prevent access of sunlight, the white ones (clothes) are best and the rest are in between the two.

——

APPENDIX IV

प्राणायामः

PRANAYAMA

प्राणायामाश्चतुर्भागो रेचको विश्रमस्तत: ।
पूरक: कुंभकश्चेति तेषु पूर्वस्य मुख्यता ।। 1

In Pranayama, of its four branches (1) Rechaka, (2) the rest (relaxation) following it, (3) Pooraka, and (4) Kumbhaka, the first one is of great importance.

निर्दिष्टो विश्रमस्तत्र बाह्यकुम्भकनामक. ।
विश्रान्तिर्दीयते ज्ञेन प्राणस्य श्वासकर्मणि ।। 2

The rest that follows exhaling (Rechaka) is known as Baahya Kumbhaka and this provides rest to the Life's activity of breathing.

मलनिस्सारणस्यैव मुख्यता प्राणकर्मसु ।
हेतुनानेन मुख्यत्वं रेचकस्यैव मन्यते ।। 3

Of the life's activities, elimination of filth is important and because this is made possible by Rechaka, it is considered important.

मलानिलबहिष्कारे रेचकेन सुसाधिते ।
श्वासाशयस्य रन्ध्रेषु वायु: शुद्धो विशेदलम् ।। 4

When the tainted air is expelled completely (almost) by Rechaka then sufficient pure air enters (in the following Pooraka) into the empty spaces of the lungs.

अंशः शुद्धिकरोवायोः प्रविश्यरुधिरागुषु ।
प्राणस्योपकरोत्यन्तः बहुधा स्वास्थ्य रक्षणे ॥ 5

The portion of the air that is cleansing in its action thus enters into the particles of the blood and aids life in eliminating filth (and other functions).

प्रविशेच्चेन्न पर्याप्ते नव्यो वायु पुनः पुनः ।
श्वासाशयस्य रन्ध्रेषु प्राणशक्तिक्षयो ध्रुवः ॥ 6

If fresh air does not enter into the air-spaces in the lungs again and again, there will surely be loss of vital power.

श्वाशे वायुरपर्याप्तो भविष्यत्यतिभोजनात् ।
मितभोजनमात्रेण पर्याप्तो ऽसौ भविष्यति ॥ 7

Eating reduces the entry of air (into the lungs) through breathing; abstemious eating alone makes it possible for the lungs to get plenty of it (air).

श्वासः प्रायेण कर्तव्यो नासाभ्यामेव सर्वदा ।
प्रारंभे रेचकः कार्यं नासाभ्यां च मुखेन च ॥ 8

As a rule one should breathe only through the nostrils. In the beginning (of the practice of Pranayama) one can do Rechaka through the nostril and the mouth also (begin with the nostril and complete with the mouth).

एकैकमपि च श्वासं दीर्घां कुर्यात् प्रयत्नतः ।
श्वाससंख्या यथा न्यूना समे काले भविष्यति ॥ 9

One should prolong each separate breath with effort, so that within the same length of time there may be fewer (but better) breaths.

निश्शेषं विसृजेद्वायुं सुदीर्घं रेचकैः कृते ।
यथा नव्यानिलः सर्वं व्याप्नुयद्वक्षसो ऽन्तरम् ॥ 10

He should expel the air from within each time by prolonged Rechakas, so that fresh air will pervade and fill the air-spaces in the chest (lungs).

स्नाता ते स्नानपूर्वं वाप्यन्यदावा यथासुखम् ।
अभ्यसेद्रेचकान् दीर्घान् श्वासाशय विशुद्धये ॥ 11

Prolonged Rechakas should be practised at the end during a bath (as may be preferred) so as to purify the lungs, (by clearing out stagnant air within).

[illegible] श्वासनाशाय सम्यगे रेचकैरलम् ।
[illegible] सम्यग्वान्तिः सद्यस्तान्त्रिकी भवेत् ॥ 12

When there is heat (inflammation) in the wind-pipe in the throat, temporary relief may be obtained by means of sufficient number of Rechakas.

शान्ति तात्कालिकीं यान्ति क्लेशा नानाप्रकारकाः ।
रेचकाभ्यासमात्रेणेत्यभ्यस्येत् तांस्तदर्थकान् ॥ 13

By the Practice of Rechaka alone a great variety of afflictions are relieved for the time being. Hence they should be practised for that end.

श्वासांगानामसामर्थ्यं तेषां सामर्थ्य सिद्धये ।
आहारनियमानुक्तान् अनुतिष्ठेत् सुधीश्चिरम् ॥ 14

He, that (suffers, because) of weakened lungs, can have lasting restoration of his lung capacity through proper dietetic righteousness set forth (in the text-book—Practical Nature-Cure or Svaadheena-Svaasthya-Mahaavidya).

यत्यतां पूरकाभ्यासः साधिते रेचके सति ।
तस्यापि वर्धयेत् कालं अभ्यासेन शनैः शनैः ॥ 15

When the practice of Rechaka has been mastered, then one may practise Pooraka (breathing in). (Also) in the practice of Pooraka the time occupied by the Pooraka should be increased steadily by practice little by little.

जठरं गुरु कुर्वीत पश्चाद्वक्षश्च पूरके ।
एवं श्वासाशये नव्यो विशेद्वायुरलं स्वयम् ॥ 16

By pushing the abdomen out and expanding the chest, the entry of fresh air into the lungs becomes spontaneous (easy and without effort).

यत्यतां कुंभकाभ्यासः साधिते पूरके सति ।
तस्यापि वर्धयेत् कालं अभ्यासेन शनैः शनैः ॥ 17

When the Pooraka is mastered then Kumbhaka (retention) may be attempted. The time by that should be steadily increased by practice.

रेचकादित्रयाणां च पृथक् कालव्यवस्थितः ।
प्रकृत्यै त्यज्यतामादौ निर्बंधो मास्तु कश्चन ॥ 18

The proportionate lengths of time for the three, namely Rechaka and the rest, should be left to Nature. There should be no violence at all.

स्वासस्यैवं सुसंस्कारः प्राणायामो निगद्यते ।
स्यादलं रोगिणां प्रायो रेचकाभ्यास एव तु ॥ 19

This practice of improved breathing is called Praanaayaama. But for patients the practice of Rechaka alone is enough. (Hence they should not, until they are sufficiently

improved (in health) take to the practice of Pooraka and Kumbhaka.

वसेच्छयीत तत्रैव गृहे वा बहिरेव वा ।
यत्र शुद्धानिलो वाति नाशुद्धे निश्चलानिले ॥ 20

There alone should one live—whether inside the house or outside—where pure air is moving, not in a place where the air is impure and stagnant.

नरकस्य समो दशो गृहान्तं निश्चलानिलः ।
तथा तमो ऽभिभूतस्य रवितेजोविवर्जितः ॥ 21

Like hell is the place inside the house, where the air is still, and which is dark for want of sunlight.

यथा वायुर्बहिः शुद्धो गृहान्तर्न तथा यतः ।
अतो गृहान्तरे वासाद् बहिर्वासो विशिष्यते ॥ 22

Air is never so pure inside a house, as it is outside, so living in the open is better than living inside a house.

वातायनान्यरुद्धानि कवाटरहितानि वा ।
पर्याप्तान्यपि चेत् सन्ति तत्र वा निवसेद् गृहे ॥ 23

If there be sufficient windows kept open, or (better still) without any shutters, then one may live in such a place inside the house.

नितरां शीतलो वायुः औषधं हि महत्तरम् ।
श्वासाशयान्तरादेयं यावच्छक्यं तदौषधम् ॥ 24

Very cold air is very good medicine, and it should be breathed into the lungs as much as possible.

शैत्यं न सहते यस्तु वपुराच्छाद्य कम्बलैः ।
नासाभ्यामेव सेवेत वायुं निर्मलशीतलम् ॥ 25

If one does not withstand cold, he should cover himself with shawls or blankets and take in the cold and pure air through the nostrils.

सहते निर्मलो देहो लाघवाच्छैत्यमुष्णताम् ।
बाध्यते समलस्ताभ्यां शक्तिमान्द्यादलाघवात् ॥ 26

The pure body would be able to withstand both cold and heat easily. But the impure body is unable to do so, because of weakness.

शीतेन वायुना रोगा जायेरन्निति धीर्भ्रमः ।
भवन्ति रोगिणो मर्त्याः हृच्यथ्यामितभोजनात् ॥ 27

The notion that diseases arise by breathing cold air is a superstition. Men become diseased by eating unwholesome foods and by over-eating.



APPENDIX V

On Applying Oil Over the Body

L. Ganesa Sarma

The question of applying oil over the body came up one of these days in the Fellowship class conducted by the I. I. N. T. in Delhi.

The usual answer given in books is, the skin is incapable of absorbing any nutrition directly and so the quality or quantity of the oil does not arise.

We study and learn quite a lot from books. But it does not mean that all that we read should be assimilated as such. There are right and wrong informations or views or partial knowledge. We must be able to pick and choose. This information about the skin is also one that is derived from books. We should accept any information only if it be convincing and had proof of fact. Where it is said that the skin is incapable of accepting any nutrition, we are given no proof of such a statement, whereas we have enough to the contrary. This has also been an agelong belief.

The skin is a covering which holds together the structure of bones and flesh. It regulates the heat of the body at a normal level, whatever be the outside temperature, be it the South Pole or the North Pole or near a furnace. While doing all this it serves also as one of the important excretary organs. It co-operates with the kidneys, whenever the latter gets weak, in compensating for its inability to eliminate the waste products mainly from protein

metabolism, namely urea and uric acid. Besides all this, it is an organ which breathes, though not as much as the lungs. The skin is not like a sheet of metal or glass; It has innumerable pores, finer than the finest cloth or sieve. Through a thin membrane like that, both gases and liquids are capable of passing through, by osmosis.

We worshippers of Mother Nature, believe that our food consists of all the Five Primary Elements, viz., Ether, Air, Light, Water and Earth. Of the five, it is a well known fact that the skin both gives out carbon dioxide and takes in oxygen. It is seen that people who live almost naked suffer less of diseases. Air-baths are strongly recommended by Natural Hygienists, as a necessity for both the recovery of Health and its maintenance. Use of artificial fabrics and other close-textured clothings are said to promote disease by shutting out air from the skin.

The whole idea of devising different methods of sun-baths, sun-basking, etc., is based on the foundamental understanding that the skin derives a lot of benefit by exposure to sunlight. The science of nutrition tells us that the skin produces vitamin D in the fat in its sub-cutaneous layer by exposure to sunlight. Children who are brought up in dark and dingy places, where sunlight is rare, suffer from softening of bones at joints, decalcification (osteomalacia) and develop bow-legs. They become a prey to rickets and tetany. This shows clearly that the skin has an important function in the nutrition of the body, which when upset or prevented causes such deep-seated diseases.

Water-Cure methods such as baths (Hot or Cold— the Hot Immersion Bath, the Hot Water Foot Bath, the

Spinal Bath, the Hip Bath, Sitz Bath etc.), the Hot and Cold fomentations, the Steam Bath, the ordinary cloth or clay packs, (cooling or stimulating) are all means of assimilating water through the skin. It may be argued that all such methods will work equally well if dry heat or cold is applied. Experience shows it is otherwise. A wet application works far better than a dry one—be it hot or cold.

It is the dryness in the atmosphere that is often found uncomfortable, the skin, especially where it is delicate, e.g., on the lips and the cheeks begin to crack and is very painful. Application of water and better still oil, relieves such conditions, making the skin supple.

There are chronic patients who do not perspire, however humid and hot the climate may happen to be. Along with dietetic reform, Sun-Baths after applying oil all over the skin improves skin health and rejuvenates the sweat glands, far quicker than a simple Sun-Bath. Even a healthy person experiences the difference of a Sun Bath with a dry skin and one with oil rubbed over the skin; in the latter case perspiration is profuse.

Burns and scars are healed without a scar if a wet bandage is applied with a piece of cloth soakad in cocoanut oil placed just next to the skin.

Skin-diseases, such as pimples, itch, boils, eczema, urticaria, psoriasis. etc., are treated by applying clay packs, wet bandages, herbal oils, Sun Baths etc.' and the patien feels immense relief from pain, irritation and burning sensations immediately on putting on the application and plenty of filthy matter is exuded with ease, enabling a rapid healing.

We advise people to apply baked lime juice on the head and body and bask in the sun for sometime before taking a Spinal Bath or a full bath, to get a cool headed feeling and silken touch of the hair. It also removes irritation in the eyes.

Patches of premature baldness had a regrowth of hair by applying crushed fresh Jabakusum flowers (செம்பரத்தம் பூ) daily an hour before the morning bath for about six months.

A boy of 15 years had plenty of grey hair and all that turned black in a few months of daily application of home-made Indian gooseberry(आमलक நெல்லிக்காய்) hair oil on coconut oil-base.

Applying pomades, vaseline and preparations of mineral base turn hair grey.

People who use hair dyes confess that they started using them for a little grey hair here and there. but very soon all the hair had turned white and they are compelled to use hair dyes regularly thereafter. There are people who had not used it for quite a long time and suddenly they desire to look young and get their hair dyed, the reaction being a swollen head and face requiring antidotes and confinement in bed for sometime.

If the skin is incapable of absorbing anything directly applied on it then (1) we should have no objection to allopaths and other druggists applying inorganic preparations on the skin and (2) they should have absolutely no effect—good or bad. We know it for certain that they often show a semblance of a cure by suppressing the eliminative process and driving the disease deep into the body and seriously affecting some vital organ later.

Can anyone achieve all the benefits seen in the various illustrative cases if the skin had not been capable of absorbing that which was applied on it? It is just because the skin is a thin, live membrane which helps osmosis (a process of interchange of fluids of different densities through a thin membrane) and the deficiency in the body nutrition is possibly made good to some extent.

One can only say this much: It is against *Vital* Economy to feed a person orally for some time if there be a fall in his health level; it is equally so to feed him through the skin. Either (1) the skin may not accept it or (2) it will cause a suppression of the disease process (which is really a progress towards health) (3) or worsen the condition, if there be not sufficient Vitality.

My views on this question need not necessarily be final. If the readers have points to supplement or contradict, let them write to the editor or to me. Let us enrich our knowledge by a study in Life's Laboratory.

Reproduced from the Jan-Febry-1964 issue of The Life Natural

INDEX

Aachamana 49
Aantarik Kumbhaka 77. 81, 82, 87, 90, 93, 106, 112
Abdominal muscles 69
Action, reflex 109
Adam's apple 41
Adenoids 107
Adults 53
Air 14, 17, 18, 29, 30,
Air, cold 33 [78, 117
Air, complimental 78, 79
Airmen 59
Air, residual 60, 70, 79
Air, supplemental 78, 79
Air, tidal 78
Allergy 49
Altitudes, high 59
Alveoli 60, 69, 70, 79
Ammonia 65
Anabolism 13
Aorta 45, 46
Arseniuretted hydrogen 65
Arterial blood pressure 69
Ashgourd 105
Asphyxiation 64
Asthma 21, 33, 49, 94, 103, 104, 110
Atmosphere, Sharing of 66
Atmospheric air 60

Baahya Kumbhaka 77, 81, 84, 87, 90, 93, 95, 112
Baby, new born 60, 53
Baths and exercises 88, 101, 106
Belly, protruberant 71
Blood 58, 66
Blood cells, red 64
Blood corpuscles 69
Blood pressure, arterial 69
Blood pressure, systemic 67
Blood, rate of pumping 59
Bodily purity 113
Bones 22
Breakfast 116, 117, 118
Breast cavity 50
Breathing, location for 33
Breathing rapid 58
Breathing through chest 86
Breathlessness 59
Bronchi 46, 58, 60, 94
Bronchial tubes 103
Bronchi, left primary 45

Bronchioles 50, 53, 58
Bronchi, right primary 45
Bronchi, secondary 45, 50
Bronchitis 94
Bronchus, left 46
Bronchus, right 46, 50

Calciferol 22
Calcium 21
Calcium phosphate 21
Cancer 34, 110
Capillaries 20, 55
Carbon-di-oxide 18, 50, 58,59,60,61,62,63,65,69
Carbon-monoxide 64
Cartilege 38
Cartilege, thyroid 41
Catabolism 13
Catarrh 108, 109
Causal relation 30
Ceilings, low 73
Central nervous system 38, 58, 71
Chest 83
Chest cavity 45
Chlorine 65, 73
Chronic 25, 104, 107, 108 110
Cigarette, slave to 34
Cilia 35
Cleansers 112
Cleansing diet 73
Cold draughts 33
Common cold 108
Common salt 18
Complimental air 78
Constipation 94, 100, 107
Consumption 110
Contractions of diaphragm 69
Convulsions 65
Coughing 38
Cross ventilation 32
Cubic space needed 66

Deafness 95
Death 64
Desire 115
Dhoti 25
Diaphragm 56, 57, 69, 70 71, 83
Diastole 74
Diet 73, 96, 100
Dietetic righteousness 31, 71, 72, 87, 96
Digestion of food 59,104
Diet, herbivorous 62
Diet, low 73
Digestion of food 59, 104
Digestive organs 19
Dioxide, sulphur 65
Dirt 35
Dress 18, 22, 23, 89
Drugs 19, 108
Dysphagia 109
Dyspnoea 59

Earache 108
Eating, reform in 71
Eczema 20,25
Edward Earl Purinton 15
Elasticity 84
Elasticity of lungs 70,71
Empyema 52
Enema 100
Enlargement of structures 46
Ether 14, 16, 17
Epiglottis 41
Eustachian tube 108
Exercise 71, 75, 118
Exhaling 83, 119
Expiration 56, 57, 67, 69 78

Fabrics, artificial 23
Fasting 73,96,99,104,118
Fat soluble vitamins 21
Filth, accumulation of 113
Five-fold food 72
Food 14, 15, 17, 70, 78
Food, gross 17
Food medicine, Natural 17
Foreign matter 38, 94
Frontal headache 109
Functions, voluntary 72

Gaayatri 121, 125
Gaayatri-Siras 121
Gases, interchange of 18, 63
Glottis 66
Goblet cells 35
Gram, Bengal 21
Gram, Green 21

Habits 71
Haemoglobin 64
Hata-yogis 116
Headache, frontal 109
Health level, lowering of 72
Health, restoration of 76
Heart 50, 53, 55, 59, 74
Heat 31
Heat regulating mechanism 23
Herbivorous diet 62
Hydro-cyanic acid 65
Hydrogen 65
Hydrogen, arseniuretted 65
Hydrogen, sulphuretted 65
Hyperpnoea 65

Indigestion 94
Inelastic 70
Inflammation 21
Inhale 78,83,101,118,119
Inorganic matter, elimination of 19
Inorganic wastes 19
Inspiration 35, 56, 66, 69
Intellect 121
Interference, surgical 49

Involuntary functions 72
Irradiation, ultraviolet 21
Itis 50, 94

Ka pasutram 113
Khaddar 24
Kidneys 19, 72
Knmbhaka 75, 76, 86, 89
Kumbhaka, aantarik 77
Kumbhaka, baahya 77, 81
Kural 16, 31

Larynx 35, 41
Leonard Williams, Dr. 30
Light 14, 17, 18
Liver 94
Lobes 55
Long life 29, 112
Lumen 58
Lungs 18, 21, 35, 41, 56, 59, 70, 72, 74, 78, 80, 83, 94, 119
Lymphatic vessels, inflammation of 109
Lymph nodes, large 46
Lymphoid tissues 107
Lymph vessels 55

Maadhyaanhikam 80
Mantras 80, 111, 112, 120, 121
Mascular activities 19
Massage 94
Mastoid bone 108
Media–stinum 55
Meditate 89, 99
Messages, flow of 71
Metabolism 13
Methane 65
Milton 13
Mind, detaching of 80
Monoxide of carbon 64
Morbid matter 49, 107
Mountain climbers 59
Mouth cancer 34
Mothers, pregnant and suckling 22
Muscle 58
Muscles, respiratory 78
Musicians, vocal 94, 95
Mucous membrane 35

Nap, midday 102
Narcotised effect 65
Nares, posterior 107, 109
Naso–pharynx 35
Nature 96, 97
Nature, law of 15
Nature-Cure 16
Nature, order of 38
Nausea 20
Nervous 70
Nervous sweat 20
Nervous system, central 38, 58, 71
Nitric-oxide 65

Nitrogen 60, 61, 65
Nitrogenous matter 62
No-breakfast plan 91
Non-violence 82, 87
Nostril 95, 107, 118, 119

Oesophagus 41 46, 55
Ointments 20
Old age, premature 15
Organs of elimination 72
Organs, resting of 74
Oxidation 31
Oxygen 18, 31, 50, 58, 60 61, 62, 63, 65
Oxygen, carrier of 64
Oxygen content 59
Oxygen, pressure exterted by 64

Pain over top of head 109
Pale 20
Passive movements 58
Peace 97
Pharynx 40, 109
Phosphorus 21
Pigmentation 21
Pitch 43
Place for Pranayama 89
Pleura 51
Pleural cavity 55
Pleura visceral 52
Pleura, parietal 52
Poisonous additives 73
Pollute 33, 74
Pooraka 69, 77, 81, 86, 87, 93, 95, 119
Positive diet 104, 105
Posterior nares 109
Posture 114
Praanavayu 113
Praanaayaama 71, 75, 76 77, 78, 79, 81, 82, 86 88, 93, 94, 95, 106, 111, 112, 114, 115, 116, 117
Praanaayaama, agarbha 119
Praanaayaama, Sagarbha 119, 120
Praatah Sandhya 80
Praayaschitta 113
Pregnant woman 22
Pressure, abdominal 57
Pressure exterd by oxygen 64
Pressure of gases 63
Pressure, blood system 67
Pressure within abdomen 56

Processing 16
Projecting teeth 43
Pronunciation 43
Proteins 62
Protruberant belly 21
Psychic shock 20
Pulmonary system 50

Purulent 52

Rechaka 69, 77, 78, 81,82 86, 87, 90, 94, 119
Rechaka through mouth 82
Rechaka through nostril 82
Red-blood cells 64
Reflex action 109
Reflex, protective 38
Reform in eating 71
Refreshed 102
Relaxation 75, 77, 98
Religious observance 80
Renunciation 97
Residual air 60, 79, 80, 82
Resipiration 60, 67
Respiration, frequency of 58
Respiratory quotient 62
Respiratory tract 35
Respiratory variations 67
Respired air 61
Ribs 55, 56
Rickets 22
Righteousness, dietetic 31, 71, 72
Rhythm of life 14
Rhythm of work and rest 15, 29, 76
Rhythm 86
Rhythmic breathing 88
Rules 87

Salts, inorganic 18
Sandhyaa-Vandana 80,111 114, 118
Sebaceous glands 20, 21
Sebum 20
Self-surrender 96
Shikaikai beans 21
Shock, psychic 20
Sinuses 40, 109
Skin 18, 23
Skin applications 19
Sleep 32, 96, 99, 100, 102
Sleeplessness 96, 97, 98
Sneezing 38
Sound 41
Space, cubic needed 66
Spasam 66
Sphenoidal Sinus 109
Spinal Bath 88, 99, 100
Sri Sankaracharya 113,114
Sternum 45, 55, 56
Stimilus, chemical 59
Subcutaneous tissue 20
Suddha Brahman 121
Suffocation 103
Sulphueretted hydrogen 65
Sun's rays 73
Supplemental air 78
Surgical interference 49
Surrender 99
Suvah 123
Svetaasvataropanishad 113, 114, 117

Swallowing, difficulty in 109
Sweating 22
Sweat, insensible 19
Sweat, nervous 20
Sweat, visible 19
Sympathetic system 20
Systemic blood pressure 67

Taittiriyopanishad 14
Tears, forcible expulsion of 109
Temperature of expired air 63
Temperature, external 19
Thoricic cavity 55, 56
Thorax 50, 53, 67, 70, 71, 83, 84
Throat cancer 34
Thyroid cartilege 41
Tidal air 78
Tiruvalluvar 16, 31
Tissues, purity of 70
Tobacco 33
Trachea 35, 43, 49, 55, 94
Tract, respiratory 35
Tuberculosis 110

Ultraviolet irradiation 21
Unconsciousness 64
Urea 18, 62
Urticaria 23

Vaccuum 78, 119
Variations, respiratory 67
Vedas 111, 123, 124
Veins 67
Vegetarian 62
Vertibral 55, 56
Ventilation 32, 64
Violence 24, 119
Violence eliminated 82
Vitality 107
Vital Economy 63, 107
Vital power 13, 14, 105
Vital reserve 15, 74
Vitamin A 15, 74
Vitamin D 21, 22
Vocal cords 41
Voluntary functions 72
Volume of expired air 61
Volume of inspired air 61
Vomitting 59
Vyaahritis 121, 122, 123, 124

Walking 107
Water 13, 17, 18, 66
Water-Cure 16
Water lost 19
Water vapour 61
Wearing, mode of 24
Windows 32
Worry 97, 106

Yogic aasanaas 106

Sarma's Sanatorium Press, Ganeshnagar, Pudukkottai--622001

ಡಾ॥ ಜಿ. ಶ್ರೀನಿವಾಸಯ್ಯನವರ ಗ್ರಂಥ ಭಂಡಾರ
ACC. No.: ಜಿ ೧೦೪೬೦೯
No.: ೦೦೧೨
DATE: ೭-೬೦೯
ಸಂದಿಸುವವರು ಪುಟ - 1

eating, besides other
of living, that is
le for the level of
gh or low. It there-
omes necessary for
e to know the Right
lealth through eating,
health or in disease.

lerstand this subject
ad :

EATING FOR HEALTH

by L. Ganesa Sarma

Rs. 2-50
ge by Regd. Re. 1-40

It is a great service to help the sickly in alleviating pain and restoring lost health. This is normally done in a sanatorium.

Your own home can also be your Health–Home

Take advantage of our :

Study Groups

Consultation Service

Write to :

L. Ganesa Sarma
Secretary
The Indian Institute Of Natural Therapeutics
Ganesanagar
Pudukkottai—622001

Or

S. Swaminathan
Duputy Director
The Indian Institute Of Natural Therapeutics
24, Outram Square
New Delhi—110001

Sarma's Sanatorium Press, Ganeshnagar, Pudukkottai--622001